Selective quadratic encodings and smaller Pell norms for the consistency of ZFC

OpenAI-assisted research for Vladimir Reshetnikov and ProveIt

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Abstract

We combine direct quadratic-system encoding with selective, rather than full Sidon, separation of monomials. Nonuniform overlapping target intervals further reduce the exponent range. A sparse signed-borrow padding replaces a dense geometric sum; a calibrated two-block binary construction then replaces a generic three-block carry-to-binomial conversion. Independently, we reduce the denominator used to absorb a Pell-index quotient and seed a square-condition guard with an already required radicand. The deliverable contains explicit division-free polynomial circuits, fixed coefficient recipes and payloads, ordinary mathematical proofs, and reproducible finite certificates. None of these equivalences proves consistency, and no global optimum or new end-to-end kernel certificate is asserted.

1 Statement, domains, and comparison boundary

The arithmetic baseline is the immutable ProveIt snapshot `124c07c0b77a`, which places eight wave-six packages as sources 38–45 beside sources 31–37. In particular, source44 is our preceding B4 delivery. The current collection is

https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

The accompanying source maps retain immutable links; later repository activity is distinguished from this frozen comparison.

For every validated equation below, the claimed theorem is

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in D^r \; F(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$

where the domain D is stated explicitly. Natural numbers include zero; positive integers do not; unrestricted integers are a third domain. Registers in the formulas are polynomial abbreviations, not additional unknowns. All powers in the final expressions have fixed nonnegative integer exponents. Square roots and divisibility appear only in proofs, not as extra operations allowed in the delivered polynomial.

The proof direction is always constructive solvability versus a finite syntactic ZFC refutation. The literal K215 calibration and native-window/Pell compiler remain identified inherited mathematical dependencies. The new adapters are ordinary proofs formalizable in ZFC. Finite audits certify source identities, exact arithmetic, degree witnesses and bindings, not unbounded compiler correctness. We neither assume nor establish Con(ZFC).

2 Results and fair comparisons

An operation is a charged binary addition, subtraction, or multiplication, including multiplication by a fixed numeral and every multiplication used for a fixed power. Shared registers and fixed integer leaves are not charged again. Degrees are total degrees in the quantified unknowns; the large coefficients are fixed. This arithmetic-circuit measure is not a bit-complexity measure or an expanded-monomial count.

Domain and unknowns	Exact degree	Operations	Comparison
$10\mathbb{Z}$	21 111 400	659	Source40: 237 520 456/782
9 positive	7 344 328	738	Source45: 129 875 980/788
9 positive, fewer gates	12 023 320	734	Same baseline
$13\mathbb{Z}$, Jones only	57 866 184	759	Source44: 226 334 238 192/1409
11 positive	958	117	Source38: 1154/123
11 positive, fewer gates	962	115	Source38: 1140/129
15 positive	80	72	Local P73: 80/73

The nine-natural versions use 744 and 740 operations, respectively, at the same degrees. The eleven-positive two-equation shifted systems have degrees $(2, 956)$ at 112 operations, or $(2, 960)$ at 110 operations. The systems are secondary presentations; the main claims above are single polynomial equations. The unseeded thirteen-integer comparator has degree 71 382 096 and 770 operations.

These rows are not ordered by one universal notion of "simplest." Sun's 2026 theorem is used only for the ten-unrestricted-integer row. The nine, thirteen, eleven, and fifteen-variable routes do not use that theorem. The thirteen-variable row is therefore a dependency-reduced alternative, not a competitor on unknown count with the ten-integer row. Positive and unrestricted domains must not be conflated.

The known 67-operation scalar equation and 59-operation system remain unbeaten on operation count alone. Our previous twelve-positive equation has smaller degree 901 than the new eleven-positive equation, at the cost of an extra unknown. Historical fifteen-positive points 99/69, 93/70, 83/71, 56/74, and 44/78 also remain relevant; the present 56/74 regrouping is explicitly not claimed as new. The eight earlier packages were reviewed as mathematical sources, not all rerun wholesale. New circuit properties are independently audited in this archive.

Coefficient cost is not hidden. All active constructions in this release use the complete source31 Wang943 121×8801 tuple: $B_{\text{native}} = 2^{1064921}$, $\beta = B_{\text{native}} - 1$, and the three matching ports K, M_C, M_F. The fourth port is already shifted. The four uncompressed payloads total 533,099 bytes. This is larger than source44's alternative 295,388-byte tuple, which is not silently substituted. The source constant $2^{2134940}$ and derived fixed leaves have exact finite recipes from this selected tuple; their cost and bit lengths are disclosed separately. The new wins are in exponent ranges, degree, arithmetic circuits, and some proof interfaces, not a claim to new coefficient-byte records.

3 What is simplified structurally

A quadratic parent with 31 positive coordinates and 22 residuals can be shifted to nonnegative coordinates without forming a quartic sum of squares. Earlier direct encodings separated all 528 padded quadratic monomials with a full B_2 set, although only 71 of those monomials occur in the residuals. We separate each occurring monomial from every possible contaminant, while allowing irrelevant monomials to collide with one another. This is a certified finite condition, not an appeal to random hashing. Target intervals may overlap provided no unwanted contribution lands on a tested coefficient.

The same principle applies to the rest of the construction: impose only the relation needed at the next interface. The binary threshold can be calibrated to the known mask instead of a generic maximum;

the absorbed index congruence needs only Y-divisibility initially, rather than full XY-divisibility. In each case the proof explicitly reconstructs what was omitted before invoking the inherited interface.

4 Selective quadratic isolation with sparse target padding

This section proves the new source encoding, independently of the number-theoretic finalizers. Its only finite input is the explicit tuple in `code_v1.json` and the 22 source residuals identified below. The proof does not assume a full Sidon (B_2) property, disjoint source intervals, or any unverified output of the search program.

4.1 The exact source and coefficient convention

Let $r_1, \dots, r_{22}$ be the quadratic residuals of the original package31 positive-variable source. Every original variable X_i is replaced by $x_i + 1$, where $x_i \in \mathbb{N} = \{0, 1, \dots\}$. We use the following coordinate order:

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma, \mathsf{N}, \mathsf{E}, \mathsf{P}, c, a,$$
$$X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

Here the displayed names label source coordinates, not the variables of a final nine- or ten-variable polynomial. In this section the coefficients may first be regarded as elements of $\mathbb{Z}[\beta, K_0, M_C, M_F]$; the final construction must substitute its specified, fixed compiler tuple. These four symbols are not additional Diophantine unknowns. The isolation identities below hold under every integer specialization of the ports.

The exact source bytes are `upstream_parent_quartic31.json`, fetched from repository commit

`124c07c0b77a9e20e84676514d8b1ff78536ea3f`.

The accompanying `SOURCE_PINS.json` records repository paths, SHA-256 hashes and Git blob hashes. The selected source is the original positive-transport version, not a silent substitution of package45's transport variant. After shifting coordinates, package45's row 15 is precisely the selected row 15 minus $\beta(x_J + 1)$. Thus its weighted constant coefficient differs by -2β, and its weighted x_J coefficient by $-\beta$; the other weighted coefficients agree. All 136 supports are the same. The checker verifies this exact difference.

4.2 A support-selective isolation theorem

For n coordinates write

$$r_k(x) = \sum_{\alpha \in S_k} p_{k,\alpha} x^\alpha, \qquad |\alpha| \le 2.$$

Choose distinct positive integer weights $w_1, \dots, w_n$ and set $w_0 = 0$. Pad each multi-index α with $2 - |\alpha|$ copies of index 0 to obtain an unordered pair. This is a bijection between multi-indices of degree at most two and the $\binom{n+2}{2}$ unordered pairs from $\{0, \dots, n\}$. Put $e(\alpha) = \sum_i \alpha_i w_i$. Choose distinct targets $q_k \in \mathbb{N}$ satisfying

$$q_k - e(\alpha) \ge 0 \qquad (\alpha \in S_k), \tag{1}$$
$$q_k - e(\alpha) + e(\gamma) = q_j \implies k = j \text{ and } \alpha = \gamma \qquad (\alpha \in S_k,\ |\gamma| \le 2). \tag{2}$$

Only the first monomial in this condition is required to occur in a residual. Distinct absent monomials are allowed to have equal weights. Define

$$f_\alpha = (2 - |\alpha|)! \prod_i \alpha_i!, \qquad c_{k,\alpha} = f_\alpha p_{k,\alpha}, \qquad D(t) = \sum_{k, \alpha \in S_k} c_{k,\alpha} t^{q_k - e(\alpha)},$$

and $C_x(t) = (1 + \sum_i x_i t^{w_i})^2$. Then, identically in all the x_i and all coefficient ports,

$$[t^{q_j}]C_x(t)D(t) = 2r_j(x). \tag{3}$$

Indeed, the contribution from monomial x^γ in the square has multinomial coefficient $2/f_\gamma$. Condition (2) permits it to reach target q_j only when $k = j$ and $\gamma = \alpha$. Its product with $f_\alpha p_{j,\alpha}$ is $2p_{j,\alpha}$. Summing proves the identity, including all signs. This argument is a polynomial identity, not a test on bounded witnesses.

The coefficient bound does not acquire a factor 22. The reversed exponents $q_k - e(\alpha)$ are injective. To see this, suppose $q_k - e(\alpha) = q_j - e(\gamma)$ with $\alpha \in S_k$ and $\gamma \in S_j$. Adding $e(\gamma)$ makes the left side reach q_j; (2) forces $k = j$ and $\alpha = \gamma$. Consequently, after fixing the integer ports, if $A = \max_{k,\alpha} |c_{k,\alpha}|$, each coefficient of the entire polynomial D has magnitude at most A. For $x_i \geq 0$,

$$\left|[t^s]C_x(t)D(t)\right| \leq A\left(1 + \sum_i x_i\right)^2. \tag{4}$$

For a fixed coefficient of the product, sum over the nonnegative coefficients of C_x and use the bound A for the single corresponding coefficient of D. Their sum is $C_x(1)$. Collisions between absent monomials in C_x, or overlaps between source intervals, do not change this argument.

Leading coefficient and positivity. Let $M = \max_i w_i$ and $Q = \max_k q_k$. In the explicit certificate the unique row at target Q is original row 5. Its shifted polynomial is

$$r_5(x) = 1 + 2x_E - x_a + x_E x_X - x_a x_X.$$

It has constant coefficient 1. Positivity of every input weight implies that the coefficient of t^Q in D is exactly 2, and no term has higher degree. If $B > A + 1$, then

$$\left|\sum_{s<Q}[t^s]D(t)B^s\right| \leq A\frac{B^Q - 1}{B - 1} < B^Q, \qquad D(B) > 0.$$

Also $D(B) < 2AB^Q$ for $B \geq 2$. Row 5 shows that the all-zero source vector is not a common zero. These facts hold independently of the values of the fixed ports.

4.3 The explicit frozen certificate

The weight vector, including its padding coordinate, is

$$(w_0, \ldots, w_{31}) = (0, 9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7,\\ 135, 38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120).$$

The row order and corresponding targets are

k	16	2	9	14	21	7	15	22	13	3	19
q_k	70	86	451	685	1050	1296	1530	1874	2212	2358	2696
k	20	1	10	17	8	4	6	11	18	12	5
q_k	2780	2792	3157	3290	3634	3929	4238	4384	4527	4865	5190

Thus

$$M = 183, \qquad Q = 5190, \qquad H = Q + 2M = 5556, \qquad (M + 1) + (H + 1) = 5741.$$

There are 71 marked padded pairs: the 32 constant/linear pairs and 39 genuine quadratic pairs. The 22 row term counts, in original row order, are

$$(4, 5, 5, 6, 5, 8, 6, 8, 8, 4, 3, 6, 9, 17, 10, 3, 6, 6, 4, 3, 4, 6),$$

with sum 136. Of all 528 padded pairs, only 267 different integer pair sums occur. For example, $w_1 + w_3 = 9 + 23 = 25 + 7 = w_2 + w_{15}$; both pairs are absent from the source. Therefore this is genuinely not a full B_2 set. The overlapping intervals, including targets only 12 apart, are intentional.

The byte-level canonical identifier for `code_v1.json` is

`53cfdccbb252fd3f6ebe87ae0bb9dad8b2fdfc8c6168f668029add30ed339e25`.

No seed, random choice, optimizer or search executable remains in the mathematical specification. A deterministic finite tuple is part of the equation, rather than an existential claim that some suitable tuple exists.

4.4 Sparse target padding: one bit absorbs incoming borrow

The same selective code permits a smaller source wrapper than dense centering. Let $B \geq 4$ be a power of two and let

$$F(t) = \sum_{s=0}^{H} a_s t^s \in \mathbb{Z}[t], \qquad |a_s| < B/4, \qquad F(B) > 0.$$

For a target set $\mathcal{Q} \subseteq \{0, \ldots, H\}$ put

$$A_{\mathcal{Q}} = \sum_{q \in \mathcal{Q}} B^q, \qquad J = 2F(B) + A_{\mathcal{Q}}.$$

Then

$$\tau_2\big(J, (B-2)A_{\mathcal{Q}}\big) = 0 \quad \Longleftrightarrow \quad a_q = 0 \quad \text{for all } q \in \mathcal{Q}, \tag{5}$$

where $\tau_2(u, v) = 0$ means that binary addition of the nonnegative integers u, v is carry-free.

For completeness, write $\delta_s = 1$ at a target and 0 elsewhere, and form the signed carry recurrence

$$c_0 = 0, \quad c_{s+1} = \left\lfloor \frac{2a_s + \delta_s + c_s}{B} \right\rfloor, \quad d_s = 2a_s + \delta_s + c_s - Bc_{s+1}.$$

Induction gives $c_s \in \{-1, 0\}$ and $0 \leq d_s < B$: the numerator lies strictly between $-B$ and B. Because $J > 0$, telescoping forces the final carry $c_{H+1} = 0$; otherwise $J = \sum_{s=0}^{H} d_s B^s - B^{H+1} < 0$. Thus the d_s are the actual base-B digits of J.

At a target the mask digit is $B - 2$, whose binary digits forbid all bits except the least significant bit. Hence carry-freeness requires $d_s \in \{0, 1\}$. If the outgoing carry were -1, the numerator would be $d_s - B \leq 1 - B$, contradicting $2a_s + 1 + c_s > -B/2 - 1 \geq 1 - B$. The outgoing carry is therefore zero. Now $2a_s = d_s - 1 - c_s$, and the only even integer allowed by $d_s \in \{0, 1\}$ and $c_s \in \{-1, 0\}$ is 0. Conversely, $a_s = 0$ gives $d_s = 1 + c_s \in \{0, 1\}$ and outgoing carry zero. This proves (5); in particular, an incoming borrow cannot produce a false zero test. No test is imposed on an absent target.

4.5 Source wrapper and size estimates before power-of-two decoding

Choose an integer $b \geq 1$ and an integer base $B > 4096Ab^2$. This assumption is a numerical coefficient majorant, not yet a power-of-two assumption. Define

$$\begin{aligned} \ell &= \sum_{i=1}^{31} B^{w_i}, & A_{\mathcal{Q}} &= \sum_{k=1}^{22} B^{q_k}, \\ N_0 &= B^{M+1}, & N_1 &= B^{H+1}, \\ N_{\mathrm{src}} &= N_0 N_1 = B^{5741}, & \mathcal{C} &= bB^M, \\ M_0 &= N_0 - 1 - (b-1)\ell, & J(g) &= 2(1+g)^2 D(B) + A_{\mathcal{Q}}, \\ S(g) &= g + N_0 J(g), & T &= M_0 + N_0(B-2)A_{\mathcal{Q}}. \end{aligned}$$

For every integer $0 \leq g < 2\mathcal{C}$, all of these expressions needed by the no-carry test are nonnegative, with $J(g), T > 0$, and

$$S(g) + T < N_{\mathrm{src}}. \tag{6}$$

Indeed, $B > 2b$ implies $g < N_0$, and the base-B digits of M_0 are $B - b$ at the distinct marked input positions and $B - 1$ elsewhere, so $0 \leq M_0 < N_0$. As g is integral, $1 + g \leq 2bB^M$. Consequently

$$0 < 2(1+g)^2 D(B) < 16Ab^2 B^H < N_1/256.$$

Furthermore, using $M \geq 1$,

$$(B-1)A_{\mathcal{Q}} \leq B^{Q+1} - 1 < N_1/B^2.$$

It follows that

$$S(g) + T < N_0\left(2 + N_1/256 + N_1/B^2\right) < N_0 N_1.$$

The last inequality follows already from $B \geq 4$ and $N_1 \geq 4$. These estimates hold before any bridge has deduced that b is a power of two. They therefore do not introduce circular reasoning into that bridge. The displayed N_{src} has no factor 4; the old dense-padding wrapper $4B^{5741}$ is not the wrapper used here.

If now b and B are powers of two, binary carry-freeness of g and M_0 is equivalent to a unique representation

$$g = \sum_{i=1}^{31} x_i B^{w_i}, \qquad 0 \leq x_i < b.$$

The mask $B - b$ allows precisely the lowest $\log_2 b$ bits in each marked cell and no bits in an unmarked cell. For this representation, $(1+g)^2 D(B) = C_x(B) D(B)$. By (4),

$$|a_s| \leq A\big(1 + 31(b-1)\big)^2 \leq A(31b)^2 < B/4.$$

The low and high words split at the power of two N_0. Combining the low-word mask, (5) and (3) proves

$$\tau_2\big(S(g), T\big) = 0 \quad \Longleftrightarrow \quad \begin{cases} g = \sum_i x_i B^{w_i}, \quad 0 \leq x_i < b, \\ r_1(x) = \cdots = r_{22}(x) = 0. \end{cases}$$

A common source zero is nonzero because $r_5(0) = 1$. Choosing b from any cofinal allowed power-of-two family larger than every x_i gives $B \leq g < \mathcal{C}$, since all weights are positive and distinct. Thus the usual cofinal source-witness requirement is preserved. The arithmetic finalizers must still prove their own bridge hypotheses, fixed coefficient specialization and final variable-domain equivalences.

4.6 Independent finite verification and limits

The standard-library program `verify_selective.py` performs the following checks without executing any upstream code or search binary:

1. Check source SHA-256 and Git blob hashes against the current pinned repository receipts, and the frozen code's exact byte hash.
2. Parse all 22 parent residual formulas, including affine abbreviations, in an independently implemented sparse polynomial ring; compare them with the supplied sparse terms.
3. Apply the 31 binomial shifts, verify every weighted coefficient in `source_support.json`, and compare all 136 terms to package45 with precisely the two row-15 corrections described above.
4. Enumerate all $136 \cdot 528 = 71\,808$ source-term/padded-pair interactions. Exactly 136 land at marked targets, each at its intended row and monomial. Reconstruct the 22 formal target polynomials and verify that each is twice its residual.
5. Check all reversed exponents are nonnegative and distinct, the leading coefficient is 2, and the displayed maxima are exact.

The replay command is `python3 verify_selective.py`; it also passes with Python's `-O` switch, because validity checks do not use removable assertions. The finite certificate supplies the complete finite premise of the isolation proof. It does not purport to prove the inherited machine compiler or the Sun/Jones number-theoretic theorems.

The maximum target decreases from the package45 value 35178 to 5190, and the maximum input weight from 799 to 183. These are source exponent spans, not final polynomial degrees or operation counts; those metrics must be computed from each compiled finalizer. The tuple is not claimed globally optimal. The construction gives no new proof of Con(ZFC), and no assertion that the consistency question has been decided. Its contribution is an explicit, rigorously checkable support-selective code and a sparse padding lemma that can be composed with the separately stated consistency-to-source theorem.

5 A selective, borrow-tolerant ten-integer equation

There is a completely fixed polynomial in ten unrestricted integer unknowns

$$F_{10}(f, g, h, k, l, w, x, y, z, m)$$

of exact total degree 21 111 400, with a division-free evaluation circuit using 659 binary operations, for which ordinary mathematics formalizable in ZFC proves

$$\neg\exists (f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The proof uses the pinned ProveIt machine/compiler calibration and Sun's arithmetic bridge. It is not an end-to-end proof-assistant certificate and it does not establish either consistency or inconsistency. The unknown count is inherited, not a new universal unknown-count theorem.

Three source-layer changes are responsible for the improvement: selective isolation of used quadratic monomials, one-bit target padding that tolerates incoming borrows, and calibration by the actual mask population instead of a uniform worst-case carry threshold. The latter two changes are proved below and are independent of any search heuristic. All scalar powers in this section have fixed nonnegative integer exponents. Auxiliary letters are polynomial abbreviations, not extra unknowns.

5.1 Fixed data and selective extraction

Use precisely the 22 quadratic residuals of `data/parent_quartic31.json` and the complete fixed port tuple

$$(\beta_{\text{native}}, K_0, M_C, M_F) = (\texttt{beta}, \texttt{K}, \texttt{MC}, \texttt{MF_source}).$$

The last port is already shifted; it must not be shifted again. Its tuple is Wang943 121×8801, with port bit lengths $(1064921, 1070004, 1064921, 1064922)$. Each payload is included, with compressed and uncompressed hashes. The active binding is `INSTANCE.json`; historical relative paths inside preserved input files are provenance only.

Replace every positive parent coordinate by $n_i + 1$. Let $r_j(n)$ denote the resulting residual, written $r_j = \sum_\alpha p_{j,\alpha} n^\alpha$, $|\alpha| \le 2$. This is the original package-31 source, not the later package-45 transport variant. The inherited source theorem is

$$\exists n \in \mathbb{N}^{31} \ \bigwedge_{j=1}^{22} r_j(n) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con}(\mathrm{ZFC}), \qquad \mathbb{N} = \{0, 1, 2, \ldots\}.$$

Its parent coordinate order is

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma, \\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\text{aux}}, \Delta).$$

Assign these coordinates the following weights, with a padding weight $a_0 = 0$:

$$(a_0, \ldots, a_{31}) = (0, 9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7, \\ 135, 38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120).$$

The residual order and corresponding targets are

$$(16, 2, 9, 14, 21, 7, 15, 22, 13, 3, 19, 20, 1, 10, 17, 8, 4, 6, 11, 18, 12, 5),$$
$$(70, 86, 451, 685, 1050, 1296, 1530, 1874, 2212, 2358, 2696, 2780, \\ 2792, 3157, 3290, 3634, 3929, 4238, 4384, 4527, 4865, 5190).$$

Write q_j for the target assigned to residual r_j, and put

$$t = 183, \quad q = 5190, \quad K_c = q + 2t = 5556, \quad E_c = t + K_c + 2 = 5741, \quad H_c = 2t + K_c + 3 + 22 = 5947.$$

For every used α and every γ with $|\gamma| \le 2$, the finite certificate verifies

$$q_j - \alpha \cdot a + \gamma \cdot a = q_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma. \tag{7}$$

All reversed exponents are nonnegative. There are 136 used source terms, 528 possible padded quadratic monomials, and $136 \cdot 528 = 71808$ checked interactions. Only 267 different pair sums occur: a full Sidon condition is deliberately not required.

Define

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}, \qquad D_c(Z) = \sum_{j,\alpha} c_{j,\alpha} Z^{q_j - \alpha \cdot a}.$$

Condition (7) implies that the 136 displayed exponents in D_c are distinct. Indeed, if two were equal, multiplying by the monomial belonging to either source term would violate isolation at its target. Thus, for $A_c \ge \max |c_{j,\alpha}|$, every coefficient of D_c has absolute value at most A_c, with no multiplicity factor. Its leading coefficient is exactly 2: the last residual is r_5 and $r_5(0) = 1$. In particular, the zero vector is not a source solution.

For $C_n(Z) = (1 + \sum_i n_i Z^{a_i})^2$, multinomial expansion gives

$$[Z^{q_j}]C_n(Z)D_c(Z) = 2r_j(n), \qquad |[Z^s]C_n D_c| \le A_c(1 + \textstyle\sum_i n_i)^2. \tag{8}$$

The factorial multiplier cancels the multinomial coefficient; isolation excludes every unwanted contribution. This proves the identity for all source vectors, not just tested examples.

Take A_c to be the maximum coefficient majorant obtained by replacing in each fixed-port recipe a monomial by $2^{\sum e_i \ell_i}$, multiplying by the absolute value of its integer coefficient, and summing. Here ℓ_i are the four verified port bit lengths. Exact integer arithmetic gives $\mathrm{bitlength}(A_c) = 2134927$ and, with

$$\kappa = 2134940, \qquad \beta_S = 2^\kappa, \qquad \lfloor \beta_S / A_c \rfloor = 16383,$$

we have

$$\beta_S > 4096 A_c \quad \text{and} \quad \beta_S > 64 A_c. \tag{9}$$

5.2 The new source polynomials

In the two unrestricted integer unknowns f, g, define

$$\begin{aligned}
b &= 1 + 3f, & B_c &= \beta_S b^2, & \mathcal{C} &= b B_c^t, \\
\ell &= \sum_{i=1}^{31} B_c^{a_i}, & N_0 &= B_c^{t+1}, & N_1 &= B_c^{K_c+1}, \qquad N = N_0 N_1, \\
A_T &= \sum_{j=1}^{22} B_c^{q_j}, & M_0 &= N_0 - 1 - (b-1)\ell, & & \\
T_c &= M_0 + (B_c - 2) A_T N_0, & J_c &= 2(1+g)^2 D_c(B_c) + A_T, & S_c &= g + J_c N_0, \\
X &= N^2 + (S_c + T_c)N + N - 1 - S_c, & & & & \\
Y &= \beta_S^{15}\, 2^{\kappa - 21}\, B_c^{5931} b. & & & &
\end{aligned}$$

Equivalently, $Y = 2^{\kappa H_c - 21} b^{11863}$; the first form avoids materializing its larger fixed coefficient. The fixed leaf $2^{\kappa-21} = 2^{2134919}$ is explicitly specified, not an unknown. Also $\mathcal{C} = 2^{\kappa t} b^{367}$ has the monomial form required by the inherited bridge.

Bounds before binary-base recovery. Suppose $f \neq 0$, b is an integer square and $0 \le g < 2\mathcal{C}$. Since $b \equiv 1 \pmod 3$, these hypotheses imply $b \ge 4$ and $f > 0$. The integer B_c is positive and even, but is not yet assumed to be a power of two. Its size implies $B_c > 2b$ and $B_c > A_c + 1$. In the following estimates write $B = B_c$ temporarily.

The base-B digits of M_0 are $B - 1$ except at the 31 designated positions, where they are $B - b$. Hence $0 \le M_0 < N_0$. The leading term of $D_c(B)$ is $2B^q$, and

$$|D_c(B) - 2B^q| \le A_c \frac{B^q - 1}{B - 1} < B^q, \qquad 0 < D_c(B) < \frac{A_c B^{q+1}}{B - 1}.$$

Because g is integral, $1 + g \le 2bB^t$. Therefore

$$0 < 2(1+g)^2 D_c(B) < \frac{8A_c b^2}{B-1} N_1 < \frac{1}{4} N_1.$$

The last inequality follows from $\beta_S > 64A_c$ and $B > 2$. Also $(B-1)A_T < B^{q+1} \le N_1/B^2$, since $t \ge 1$. Consequently

$$0 < J_c + (B-2)A_T < \tfrac{1}{2} N_1.$$

We have $g < N_0$, so $g + M_0 < 2N_0$. As $N_1 \geq 4$, this proves

$$0 \leq S_c, \quad 0 \leq T_c, \qquad S_c + T_c < N. \tag{10}$$

In particular $X \geq N^2 > 3b$ and $Y > \max(b, 256)$. Furthermore $B_c \equiv N_0 \equiv N \equiv 1 \pmod 3$, $M_0 \equiv 0$ and $T_c \equiv -22$, giving

$$X \equiv 1 + T_c \equiv 0 \pmod 3.$$

These statements were obtained without assuming that a square b is a power of two. This order matters in the reverse bridge.

5.3 Sparse padding that tolerates borrows

Now assume additionally that b is a power of two. Then B_c and N_0 are powers of two. The condition $g \,\&\, M_0 = 0$ and $0 \leq g < N_0$ is exactly

$$g = \sum_{i=1}^{31} n_i B_c^{a_i}, \qquad 0 \leq n_i < b.$$

The weights are distinct, so the tuple is unique. Indeed, the low mask forbids every bit outside the low $\log_2 b$ bits of these designated base-B_c cells.

For this tuple let $h_s = [Z^s]C_n D_c$ and let ϵ_s be the indicator of the 22 target positions. Equations (8) and (9) give $|h_s| < B_c/4$, and

$$J_c = \sum_{s=0}^{K_c} (2h_s + \epsilon_s) B_c^s.$$

This is a signed-coefficient expansion, not yet a digit expansion. Set $u_0 = 0$ and carry out ordinary Euclidean digit normalization:

$$u_{s+1} = \left\lfloor \frac{2h_s + \epsilon_s + u_s}{B_c} \right\rfloor, \qquad d_s = 2h_s + \epsilon_s + u_s - B_c u_{s+1}.$$

Induction gives $u_s \in \{-1, 0\}$ and $0 \leq d_s < B_c$. At a marked position $\epsilon_s = 1$, the binary mask $B_c - 2$ permits exactly the digits 0 and 1. If $h_s = 0$, the unnormalized digit is $1 + u_s \in \{0, 1\}$, so it passes. If $h_s \geq 1$, that digit lies in $[2, B_c)$ and fails. If $h_s \leq -1$, it lies strictly between $-B_c + 1$ and 0, so its normalized digit is at least 2 and again fails. Thus each marked test detects precisely $h_s = 0$, regardless of an incoming borrow. Positivity of J_c was already proved, so these are ordinary digits of a nonnegative integer.

Concatenation at the power-of-two boundary N_0 now yields

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B_c^{a_i},\ 0 \leq n_i < b, \quad r_1(n) = \cdots = r_{22}(n) = 0. \tag{11}$$

The dense geometric padding used by the previous construction has been replaced by only one extra bit at each marked cell. Unmarked negative coefficients may generate borrows; the proof allows them rather than silently assuming them absent.

5.4 Calibrating the binomial threshold by the mask population

For $N = 2^u$ and nonnegative integers S, T satisfying $S + T < N$, set

$$X_* = N^2 + (S + T)N + N - 1 - S.$$

Its leading, middle, and low blocks are disjoint. If s_2 denotes binary population, then

$$\begin{aligned} s_2(X_*) &= 1 + s_2(S+T) + u - s_2(S) \\ &= 1 + u + s_2(T) - \big(s_2(S) + s_2(T) - s_2(S+T)\big). \end{aligned}$$

The parenthesized term is nonnegative and vanishes exactly when $S \,\&\, T = 0$. Legendre's formula gives $\nu_2\binom{2X_*}{X_*} = s_2(X_*)$. Consequently

$$2^{1+u+s_2(T)} \mid \binom{2X_*}{X_*} \quad \Longleftrightarrow \quad S \,\&\, T = 0. \tag{12}$$

The stronger range $S + T < N$ is essential and was proved in (10); separate inequalities $S < N, T < N$ would not suffice.

Write $b = 2^k$ and $B_c = 2^v$ on the binary branch. The low mask has population $(t+1)v - 31k$, while each of the 22 high marked cells contributes $v - 1$ bits. Hence

$$s_2(T_c) = (t + 1 + 22)v - 31k - 22, \qquad u = E_c v,$$

and

$$1 + u + s_2(T_c) = H_c v - 31k - 21 = \kappa H_c + (2H_c - 31)k - 21 = \log_2 Y.$$

Combining (12) and (11) proves the required reverse source implication. Conversely, a source solution is nonzero because $r_5(0) = 1$. Choose arbitrarily large $b = 4^s$ exceeding all its coordinates and put $f = (b-1)/3$. Its encoding satisfies $b \le B_c \le g < \mathcal{C}$ and all carry tests pass. Thus it supplies arbitrarily large forward witnesses with $Y \mid \binom{2X}{X}$. This establishes the entire source contract for the unchanged finalizer.

5.5 Explicit radical-free finalizer

The following formulas are the inherited package-31 quintic, merged-square, structural-weight finalizer. They are written out to fix the actual equation. Distinguish its local B_*, S_*, T_* from the source base and masks above. Define

$$\begin{gathered} L = lY, \qquad U = 2LX, \qquad V = 4gwY, \qquad W = bw, \\ A = U(V+1), \qquad B_* = 2X + 1, \qquad C = B_* + (A-2)h, \\ P = U^2 V, \qquad K = X + 1 + k(P-2), \\ S_* = 2A - 5, \qquad T_* = 3WC - 2(W^2 - 1), \\ R_2 = (P^2 - 4)K^2 + 4, \\ \Delta = 8\mathcal{C}^3 g K^2 - g^2\{32(C - KL)^2\mathcal{C}^3 + g^2 K^2\}, \\ O = f^2 l^2 \Delta, \qquad \xi = zx, \\ D = (A^2 - 4)C^2 + 4, \qquad E = C^2 D\xi, \\ F = 4(A^2 - 4)E^2 + 1, \qquad G_* = 2(A+2)(A-2)^2 E^2 - 1, \\ \Omega = 16E, \qquad H_* = C + (B_* - C)F + \Omega F y, \\ I_* = (G_*^2 - 1)H_*^2 + 1. \end{gathered}$$

Put

$$\begin{aligned}
r &= T_* + S_* y, & \Sigma &= \xi S_*^5, & & \\
\Theta &= (r^3 - rS_*^2 - S_*^3)(r^2 + 23S_*^2), & & & & \\
a_0 &= b\{1 + (3O-4)b^2z^2\}, & a_1 &= R_2, & a_2 &= DFI_*, \\
w_0 &= 1, & w_1 &= a_0, & w_2 &= a_0(P^2 + K^2 + 3), \\
Z &= \Sigma m - \Theta, & A_i &= \Sigma^2 w_i^2 a_i \quad (0 \le i \le 2), & & \\
V_n &= Z^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z^2A_2 - 4A_0A_1. & &
\end{aligned}$$

The promised single equation is

$$\boxed{F_{10} = W_n^2 - 16V_n^2Z^2A_2 = 0.} \tag{13}$$

Every expression is an integer polynomial. In particular, radicals and binomial coefficients appear in the proof only, never in the equation.

Exact dependency boundary. Sun's 2021 Theorems 4.12–4.13, specialized as in his 2026 Propositions 6.1–6.2, provide the two bridge directions for any source satisfying the contract just proved. Forward witnesses have b square, R_2 square, $S_* \mid T_*$, $C = u_{B_*}(A)$ and $O > 0$. Reverse witnesses with b square, $O > 0$, $\xi \neq 0$, R_2, DFI_* square and $S_* \mid T_*$ recover $b \in 2\uparrow$, $0 < g < 2\mathcal{C}$ and the required binomial divisibility. The residue interpolation theorem permits ξ to be divisible by any prescribed positive integer and permits any residue class for $y \pmod{\xi}$. The only data substituted here are the newly proved source polynomials; the bridge does not require $Y = N^2$.

For completeness, the inherited finalizer implications can be verified as follows. The polynomial $(v^3 - v - 1)(v^2 + 23)$ has a root modulo every positive integer and none over $\mathbb{Q}$. For odd primes other than 23 the cubic discriminant is -23: if the cubic is irreducible its Frobenius permutation is even, so the quadratic has a simple root; otherwise the cubic has a simple root. At 23 the cubic has simple root 3. At 2, $-23 \equiv 1 \pmod 8$. Hensel lifting and the Chinese remainder theorem finish the local-root argument. Prime valuations in the two homogeneous factors show that $\Sigma \mid \Theta$ forces $S_* \mid T_*$: if $v_p(S_*) > v_p(T_*)$, the valuation of Θ is $5v_p(T_*) < 5v_p(S_*)$. Since S_* is odd, it never vanishes; the absence of rational roots ensures $\Theta \neq 0$.

When $z \neq 0$, a_0 square implies that its coprime factors b and $1 + (3O-4)b^2z^2$ are signed squares. The congruence $b \equiv 1 \pmod 3$ excludes a negative square and excludes zero, including the zero-factor exception. Thus b is a positive square and the usual positivity Pell lemma gives $O > 0$. Conversely, forward witnesses have $O \ge 4$: the positive integer Δ is divisible by $g \ge b \ge 4$. For $d = 3O - 4 > 0$, $d \equiv 2 \pmod 3$ is nonsquare. A positive Pell unit raised to its order modulo b gives $v = bz > 0$ with $u^2 - dv^2 = 1$, making a_0 a square. Residue interpolation then supplies $\xi = zx$ and a y that simultaneously makes a_2 square and $\Sigma \mid \Theta$.

If $z \neq 0$, the displayed a_0 always has $|a_0| > 1$. The exceptional possibilities $a_0 = 0, \pm 1$ force $|b| = 1$, hence $b = 1$, $f = O = 0$, and $a_0 = 1 - 4z^2$, a contradiction. Also $\sqrt{|a_1|} \le P^2 + K^2 + 2$. Thus the weights obey

$$|w_1| > \sqrt{|a_0|}, \qquad |w_2| > \sqrt{|a_0|} + |w_1|\sqrt{|a_1|}.$$

In the multiquadratic field, an automorphism changing a nonzero weighted root has a largest changed index; its magnitude exceeds the sum of all smaller changed magnitudes. A rational weighted sum can therefore occur only when every a_i is an integer square. This also covers negative radicands by complex absolute values, and zero radicands separately.

The exact norm identity

$$F_{10} = \prod_{\epsilon \in \{\pm 1\}^3} \left(Z + \sum_{i=0}^{2} \epsilon_i \sqrt{A_i} \right)$$

has 35 expanded terms and is checked independently by integer polynomial arithmetic. If $\Sigma = 0$ then $F_{10} = \Theta^8 \neq 0$. At a zero of F_{10} we therefore have $\Sigma \neq 0$, hence $z \neq 0$; separation forces all a_i square and $\Sigma \mid \Theta$. The reverse bridge and source proof recover a parent solution. Conversely the forward construction just described makes all a_i square and $\Sigma \mid \Theta$; choosing m to cancel one signed factor gives $F_{10} = 0$. Composition with the pinned parent theorem proves the stated consistency equivalence in ZFC.

5.6 Exact finite certificates and comparison

The literal DAG `generated/signed10_b2.json` has ten variable leaves, 18 fixed leaves, and 659 live binary arithmetic gates: 214 additions, 26 subtractions and 419 multiplications. Multiplication by a numeral is charged; fixed-coefficient construction is outside this arithmetic-circuit cost. It is not a bit-complexity claim.

The independent checker does not import or execute the builder. It reconstructs the shifted quadratic source and fixed recipes, verifies all selective interactions, checks 52 local polynomial circuit identities exactly, checks the norm identity, and audits liveness and the gate ledger. For the total degree it propagates a syntactic upper bound, then tracks the coefficient of that degree after substituting the same indeterminate for all ten variables. The resulting nonzero residues are

p	101	1009	1000003
$[T^{21111400}]F_{10}(T,\dots,T) \pmod p$	36	42	374910.

Any one of these gives the exact lower bound, hence the total degree is exactly 21 111 400. This is not a random evaluation argument. Finite small-base audits additionally corroborate the borrow and population lemmas, including non-power-of-two square source bases. They are tests of implementations, not substitutes for the unbounded proofs.

The previous package-40 ten-integer point was degree 237 520 456 with 782 operations. The new degree is lower by a factor exceeding 11.25, and the circuit uses 123 fewer operations, a reduction of about 15.7%. Package 43's 809-operation variant is also improved. These are comparisons of explicit constructions at the pinned repository snapshot, not claims of global optimality, of an absolute "simplest" equation, or of improvement over different variable/domain tradeoffs.

The pinned revision is `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. `SOURCE_PINS.json` records the verified exact source bytes. Primary arithmetic references are Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Sci. China Math. 64 (2021), 281–306, https://arxiv.org/html/1704.03504v7; and *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, https://arxiv.org/html/2609.01594v2. The inherited finalizer proof and coefficient adapter are preserved in `upstream/`. Their original relative links describe upstream layouts; the source pin manifest provides their current verified origins.

6 Nine unknowns through a parity-calibrated carry block

The explicit polynomial below has nine nonnegative integer unknowns, exact degree 7 344 328, and 744 binary arithmetic operations. Its positive-coordinate translate has the same degree and 738 operations (211 additions, 26 subtractions, 501 multiplications). All intermediate names denote polynomial registers, not quantified variables. With the identified inherited compiler and Jones ratio theorems, ordinary mathematics formalizable in ZFC proves

$$\neg\exists \mathbf{x} \in \mathbb{N}^9 \; F_9(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This is not a proof of consistency, a claim of minimum complexity, an unrestricted-integer nine-variable result, or a new end-to-end kernel proof.

6.1 Fixed data and selective extraction

The source is the original package31 system of 22 quadratic residuals in 31 positive coordinates, followed by the shift $y_i = x_i + 1$. Its exact JSON is `upstream/upstream_parent_quartic31.json`, SHA256 beginning `1c336ba0b3c9827f`; the full digest and repository pin `124c07c0b77a` are in `upstream/SOURCE_PINS.json`. We do not replace its transport row by the distinct package45 normalization. The whole 121-by-8801, 943-tile compiler tuple is bound in `COEFFICIENT_BINDING.json`; its four port bit lengths are $(1064921, 1070004, 1064921, 1064922)$. The M_F port is the already shifted $M_{F,\mathrm{source}}$. All four authenticated compressed little-endian payloads and the complete tuple adapter proof are included.

For the shifted residual p_j, set

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i! \, [\mathbf{x}^\alpha] p_j.$$

Exact source expansion produces 136 terms and 26 distinct recipes, including literal-only recipes. From the port bit lengths, termwise triangle bounds prove $|c_{j,\alpha}| < M = 2^{2134927}$. Fix

$$\kappa = 2134940, \qquad a = 2^\kappa = 8192M, \qquad c = 2^{15\kappa/2-10} = 2^{16012040}.$$

These are fixed ordinary integers. The evenness of κ will matter.

The 31 weights, in the source variable order, are

$$9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7, 135,$$
$$38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120.$$

The row order and corresponding targets are

$$16, 2, 9, 14, 21, 7, 15, 22, 13, 3, 19, 20, 1, 10, 17, 8, 4, 6, 11, 18, 12, 5;$$
$$70, 86, 451, 685, 1050, 1296, 1530, 1874, 2212, 2358, 2696, 2780,$$
$$2792, 3157, 3290, 3634, 3929, 4238, 4384, 4527, 4865, 5190.$$

Write $W = 183$, $Q_c = 5190$, $K_c = Q_c + 2W = 5556$, and $E_c = K_c + W + 2 = 5741$. The frozen finite certificate verifies, for every source-supported α and every $|\beta| \le 2$,

$$q_j - \alpha \cdot w + \beta \cdot w = q_l \quad \Longleftrightarrow \quad j = l \text{ and } \alpha = \beta.$$

It checks all $136 \cdot 528 = 71808$ interactions. Only the 71 used padded pairs need isolation; absent pair sums may coincide. The reversed exponents are nonnegative and mutually distinct, so no coefficient inflation factor occurs. Define

$$D_c(t) = \sum_{j,\alpha} c_{j,\alpha} t^{q_j - \alpha \cdot w}, \qquad C_{\mathbf{x}}(t) = \left(1 + \sum_i x_i t^{w_i}\right)^2.$$

Factorial cancellation gives $[t^{q_j}](C_{\mathbf{x}} D_c) = 2p_j(\mathbf{x})$. The highest coefficient of D_c is 2 at exponent 5190: residual5 is last and has constant term 1. Thus, for $B - 1 > M$,

$$B^{5190} < D_c(B) < \frac{MB^{5191}}{B-1}.$$

6.2 Sparse target padding and source bounds

Use $(\epsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0, n) \in \mathbb{N}^9$, where $\mathbb{N} = \{0, 1, \ldots\}$. Put $g = g_0 + 1, h = h_0 + 1, s = s_0 + 1, w = w_0 + 1, i = i_0 + 1, j = j_0 + 1$, and define

$$\begin{aligned}
b &= \epsilon + 2, & B &= ab^2, & \ell &= \sum_i B^{w_i}, \quad A_t = \sum_j B^{q_j},\\
N_0 &= B^{184}, & Z &= B^{5557}, & L &= 2bB^{5741}, \quad N = cB^{2966},\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2(1+g)^2 D_c(B) + A_t,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)A_t,\\
R &= L^2 + (S_c + T_c)L + L - 1 - S_c, & G_{\mathrm{cap}} &= 32bB^{183}.
\end{aligned}$$

The block length L and ratio threshold root N are different. On every outer assignment, $J_c, S_c > 0$, and the base-B digits of M_c are $B - b$ at the weights and $B - 1$ elsewhere through W. Hence $0 \le M_c < N_0$, $T_c \ge 0$, and

$$R = L^2 + (L-1)S_c + LT_c + L - 1 \ge 8, \qquad 0 < b \le R, N, \quad N \ge 8, \quad G_{\mathrm{cap}} > 0.$$

These facts do not presuppose a power-of-two base or a bound on g.

If $g < 4bB^{183}$, integrality yields $1 + g \le 4bB^{183}$ and $g < N_0$. Therefore

$$2(1+g)^2 D_c(B) < \frac{32Mb^2}{B-1} Z = \frac{B}{256(B-1)} Z < \frac{Z}{128}, \qquad A_t < \frac{Z}{128}.$$

Thus $0 < J_c < Z/64 < Z$, while $(B-2)A_t < B^{5191} \le Z$. Using integrality again gives

$$0 \le S_c, T_c < N_0 Z = B^{5741}, \qquad S_c + T_c < 2B^{5741} < L.$$

Once b is a power of two, so is B. Binary disjointness of g and M_c is exactly $g = \sum_i x_i B^{w_i}$ with $0 \le x_i < b$. Let $h_r = [t^r](C_{\mathbf{x}} D_c)$. Then

$$|h_r| < M(1 + \sum x_i)^2 < 1024Mb^2 = B/8, \qquad J_c = \sum_{r=0}^{5556} (2h_r + \delta_r) B^r,$$

where δ_r is 1 at a target and 0 elsewhere. In signed-digit normalization the incoming borrow is always -1 or 0: the coefficient bound preserves this by induction. At a target the normalized digit is $(2h_r + 1 + \text{borrow}) \bmod B$. It avoids the mask $B - 2$ precisely when $h_r = 0$: for zero it is 0 or 1; for positive h_r it is at least 2 and below $B/2$; for negative h_r it is above $B/2$. Consequently

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ codes a common nonnegative zero of all } p_j.$$

The symbol & denotes ordinary bitwise intersection of nonnegative integers. It occurs only in the proof, not in the polynomial circuit.

6.3 The new block-parity calibration

For $S, T \ge 0$, $S + T < L = 2^u$, the polynomial

$$X = L^2 + (S+T)L + (L - 1 - S)$$

has disjoint top, middle and bottom binary blocks. Thus

$$\mathrm{pop}(X) = u + \mathrm{pop}(T) + 1 - \big[\mathrm{pop}(S) + \mathrm{pop}(T) - \mathrm{pop}(S+T)\big].$$

The bracket is nonnegative and vanishes exactly if $S\&T = 0$. Legendre's formula gives $v_2\binom{2X}{X} = \mathrm{pop}(X)$, so

$$2^{u+\mathrm{pop}(T)+1} \mid \binom{2X}{X} \quad \Longleftrightarrow \quad S \,\&\, T = 0.$$

Put $b = 2^k, B = 2^v$, with $v = \kappa + 2k$. The low mask contributes $184v - 31k$ bits; the 22 high target digits contribute $22(v-1)$. Hence

$$\mathrm{pop}(T_c) = 206v - 31k - 22.$$

Choosing $L = 2bB^{5741}$ gives $u = 1 + k + 5741v$, and therefore

$$\begin{aligned} u + \mathrm{pop}(T_c) + 1 &= 5947v - 30k - 20 \\ &= 5947\kappa + 11864k - 20 \\ &= 2\big(15\kappa/2 - 10 + 2966v\big) = 2\log_2 N. \end{aligned}$$

Thus the tested threshold is exactly N^2, even with b linear in the input. No extra unknown, dummy digit or square-base restriction is needed. This is the new parity-calibration step.

Conversely, for a source zero choose any sufficiently large power of two $b > \max x_i$, and $\epsilon = b - 2$. Since $p_5(0) = 1$, its code g is positive. Moreover $g \le (b-1)\ell < bB^{183}$, so $32g < G_{\mathrm{cap}}$. Both masks are disjoint and $N^2 \mid \binom{2R}{R}$. Every sufficiently large power of two is available, eliminating any hidden cofinality assumption.

6.4 Ratio registers and the near-square finalizer

Use the inherited Jones ratio registers

$$\begin{gathered} U = N^2 w, \quad Y = N^2 s, \quad M_0 = RY, \quad P = 2M_0^2 U, \\ K = R + 1 + h(P-1), \quad A = M_0(U+1), \quad B_1 = 2R+1, \quad W_1 = bw, \\ C = B_1 + W_1 + \phi, \quad J = 4A - 5, \quad Q = 3W_1 C - 2(W_1^2 - 1), \\ D = (A^2-1)C^2 + 1, \quad E = iJDC^2, \quad F = (A^2-1)E^2 + 1, \\ G = A + F(CD + 1 - A), \quad H = B_1 + jC, \quad I = (G^2-1)H^2 + 1, \\ a_0 = (P^2-1)K^2 + 1, \quad a_1 = 64DFI, \quad u_0 = JF, \quad v_0 = QF + J(H - C), \\ \mu = K^2 G_{\mathrm{cap}} - 4(C - KY)^2 G_{\mathrm{cap}} - 8gK^2. \end{gathered}$$

Here u_0, v_0 avoid conflict with the binary block exponent. The imported `ratio_sound` and `ratio_complete` in `upstream/NineRatio.lean` relate the four conditions

$$a_0 \text{ square}, \quad DFI \text{ square}, \quad u_0 \mid v_0, \quad \mu > 0$$

to the central-binomial divisibility and b a power of two. Soundness gives $8g < G_{\mathrm{cap}}$; completeness requires $32g < G_{\mathrm{cap}}$. All numerical hypotheses were proved unconditionally above. Multiplying DFI by 64 does not change its square predicate.

For every outer assignment, $A \ge 2, J \ge 3, D \ge A^2, K > 1$, all Pell registers are positive, and $Q, H - C > 0$. Thus

$$0 < \frac{v_0}{u_0} = \frac{Q}{J} + \frac{H - C}{F} \le Q + H - C =: M_d.$$

Put

$$V_0 = PK, \quad t_0 = iJC, \quad S_0 = t_0(8D^2 - 4D - 1), \quad V_1 = S_0 GH.$$

As $E = t_0 DC$, exact identities yield

$$DF = t_0^2 D^3(D-1) + D, \qquad S_0^2 - 64DF = t_0^2(8D+1) - 64D \geq 8D + 9 > 0.$$

Since $I \leq G^2H^2$ and $V_0^2 - a_0 = K^2 - 1 > 0$, we have $0 < a_r < V_r^2$ for $r = 0, 1$. Define the requested single polynomial by

$$\Omega = u_0(2\mu - 1), \qquad \Theta = u_0 n + v_0 - \Omega(M_d + V_0 V_1),$$
$$\boxed{F_9 = \left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1.}$$

The four-factor identity implies, at any zero, $\Theta/\Omega = r_0 + V_0 r_1$ with $r_i^2 = a_i$. This rational number cannot be zero, because then $|r_1| = |r_0|/V_0 < 1$, contrary to the positive integer a_1. The identity $2(\Theta/\Omega)r_0 = (\Theta/\Omega)^2 + a_0 - V_0^2 a_1$ forces both roots rational and thus integral. This argument does not assume independence of square classes. The strict bounds give

$$Z_0 = M_d + V_0 V_1 + r_0 + V_0 r_1 \geq M_d + 1 > 0.$$

The factor equation is $n + v_0/u_0 = (2\mu - 1)Z_0$. Its positive left side forces $\mu > 0$, and its integral right side forces $u_0 \mid v_0$. Conversely the four ratio conditions yield an integer $n = (2\mu - 1)Z_0 - v_0/u_0 \geq 1$ and hence a zero. This completes both directions of the stated reduction.

6.5 Exact degree and audited implementation

A raw degree walk would miss a leading cancellation. The certificate uses

$$\begin{aligned} \Delta_V = V_1^2 - a_1 &= [t_0^2(8D+1) - 64D]G^2H^2 + 64DF(H^2 - 1), \\ E_0 &= u_0 n + v_0 - \Omega M_d, \\ \Theta^2 - \Omega^2(a_0 + V_0^2 a_1) &= E_0^2 - 2E_0\Omega V_0 V_1 + \Omega^2 V_0^2 \Delta_V - \Omega^2 a_0. \end{aligned}$$

The first summand of Δ_V has degree 1 763 760 and leader $2^{147} a^{881800} c^{96}$. The third summand of the last display dominates, with degree 3 672 164 and leader $2^{299} a^{1835914} c^{224}$. Finally $4\Omega^4 V_0^2 a_0 a_1$ has degree 5 998 540, strictly less than twice the dominant degree. Thus

$$\deg F_9 = 7\,344\,328, \qquad \mathrm{LC}_{\mathrm{diagonal}}(F_9) = 2^{598} a^{3671828} c^{448} > 0.$$

The diagonal attains the full multivariate total-degree bound. The cancellation-resolved certificate records every decisive exact leader; coefficient-dependent lower heads are left unknown, not guessed.

Replacing every natural input by a positive input minus one is a bijection and preserves degree. In particular $b = \epsilon_+ + 1$, $\phi = \phi_+ - 1$, and $n = n_+ - 1$. The conservative three-block comparison has degree 12 023 320 and 734 positive operations. A square-base uncalibrated two-block comparison has degree 14 687 132 and 740 operations. Its admissible bases $b = 2^{2r+1}$ are explicitly cofinal. Both comparison circuits and their proof details are included; the parity-calibrated version is primary.

The self-contained standard-library scripts `build_nine.py` and `verify_nine.py` rebuild all 12 circuits, authenticate the full tuple, verify all 26 recipe bounds and the selective certificate, and perform 360 modular replays covering 11520 independently computed named registers. They also check 43945 two-block cases, 42 threshold identities, an exact symbolic near-square identity, 243 majorant fixtures and 1728 finalizer fixtures. These finite checks audit data and implementation, not consistency. The complete proof and all limitations are in `PROOF.md`; provenance and imported dependencies are distinguished from the new mathematics.

7 Thirteen integers: a seeded guard stack and a calibrated carry block

The polynomial specified below has exactly thirteen unrestricted integer unknowns, exact total degree 57 866 184, and a division-free circuit of 759 live binary operations (520 multiplications, 215 additions and 24 subtractions). Its integer zero existence is equivalent to a finite syntactic ZFC refutation. Thus, with the stated inherited compiler theorem, ZFC proves that its lack of integer zeros is equivalent to Con(ZFC). This is an ordinary mathematical proof, not a new Lean certificate, a proof of consistency, or a global minimality claim. Unlike the ten-integer companion, the arithmetic finalizer here uses only the established Jones ratio interface, not Sun's 2026 bridge.

7.1 A coherent fixed source and an explicit polynomial

Let $p_1, \dots, p_{22}$ be the original package-31 quadratic residuals after substituting $x_i = n_i + 1$ in all 31 positive source coordinates. The exact source is `upstream/parent_quartic31.json`; the complete fixed port mapping is $(\beta, K_0, M_C, M_F) = (\beta, K, M_C, M_{F,\mathrm{source}})$ from `FIXED_BINDING.json`. All four compressed little-endian payloads are included and hash-authenticated. This uses the original transport coordinate, not package 45's extra transport shift. The imported source theorem says

$$\exists n \in \mathbb{N}^{31} \ \bigwedge_{j=1}^{22} p_j(n) = 0 \quad \Longleftrightarrow \quad \text{there is a finite ZFC refutation.}$$

The fifth residual equals 1 at $n = 0$, so any common zero is nonzero.

Use the selective weights w_i, row order, and targets q_j in `CODE.json`, identical to the companion's certified tuple. Their maxima are $W = 183$ and $Q_* = 5190$; put $K_* = Q_* + 2W = 5556$ and $E_* = K_* + W + 2 = 5741$. For $p_j(n) = \sum_\alpha p_{j,\alpha} n^\alpha$, set

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}, \qquad D_c(B) = \sum_{j,\alpha} c_{j,\alpha} B^{q_j - \alpha \cdot w}.$$

The finite isolation certificate checks all 71 808 possible interactions of the 136 used source terms with the 528 padded quadratic monomials. It proves

$$[t^{q_j}]\Big(1 + \sum_i n_i t^{w_i}\Big)^2 D_c(t) = 2p_j(n).$$

It also proves that the displayed exponents in D_c are distinct and its leading coefficient is 2. All $|c_{j,\alpha}| < A_* = 2^{2134927}$; the verifier derives this bound from the actual fixed-port bit lengths.

The unknowns are

$$(\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, r, v) \in \mathbb{Z}^{13}.$$

Every subsequent symbol is a polynomial register, not an additional unknown. Fix

$$\kappa = 2134940, \quad a = 2^\kappa = 8192 A_*, \quad \eta = 2^{16012040} = 2^{15\kappa/2 - 10},$$

and define

$$
\begin{aligned}
b &= 2\epsilon^2, & B &= ab^2,\\
\ell &= \sum_i B^{w_i}, & T_* &= \sum_j B^{q_j},\\
N_0 &= B^{W+1}, & L &= 2bB^{E_*},\\
N &= \eta B^{2966}, & c &= bB^W,\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2(1+g)^2 D_c(B) + T_*,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)T_*,\\
R &= L^2 + (S_c + T_c)L + (L - 1 - S_c), & G_c &= 32c.
\end{aligned}
$$

All powers have fixed nonnegative integer exponents. In particular, N is an ordinary polynomial, not a variable-exponent abbreviation.

Keep the literal Jones registers, with $\omega = w^2$:

$$
\begin{aligned}
U &= N^2\omega, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R+1, & W_1 &= b\omega,\\
C &= B_1 + W_1 + \phi, & J &= 4A - 5, & Q &= 3W_1C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= B_1 + jC, & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
u &= JF, & t &= QF + J(H - C).
\end{aligned}
$$

Set

$$\delta = C - KY, \qquad \Delta = g(c-g)K^2 - c^2\delta^2, \qquad (d_1, d_2, d_3) = (3s-1, 3\phi+2, 3\Delta-1).$$

The new seeded stack is

$$C_0 = a_0, \qquad B_j^* = 1 + 4d_jC_{j-1}z_j^2, \qquad C_j = C_{j-1}B_j^* \quad (1 \le j \le 3), \qquad C_* = C_3.$$

Finally define

$$
\begin{aligned}
Z &= \epsilon w i z_1 z_2 z_3 a_0, & \Sigma &= Zu^5, & \rho &= t + ur,\\
\Theta &= (\rho^3 - u^2\rho - u^3)(\rho^2 + 23u^2), & X &= \Sigma v - \Theta,\\
A_0 &= \Sigma^2 C_*, & A_1 &= \Sigma^2 C_*^2 a_1.
\end{aligned}
$$

The requested thirteen-integer equation is the following single polynomial:

$$\boxed{\mathcal{F}_{13} = (X^2 - A_0 - A_1)^2 - 4A_0A_1 = 0.} \tag{14}$$

The literal division-free DAG is `generated/signed_seeded.json`.

7.2 Sparse padding and unconditional size estimates

Assume first only $b \ge 2$. Since $B - 1 > A_*$ and the top coefficient of D_c is 2, we have $D_c(B) > B^{Q_*} > 0$. Also $0 < M_c < N_0$. For every $g \ge 1$, it follows that $S_c, T_c > 0$ and

$$R = L^2 + S_c(L-1) + T_cL + L - 1 > L^2.$$

Moreover

$$N^2 = 2^{-20}a^{15}B^{5932} < B^{5947} < B^{11482} < L^2, \qquad b \le c \le N.$$

Thus $R \ge N \ge 8$ and $b \le N, R$ before assuming any carry or square condition. Negative powers here occur only in this comparison proof, never in the equation.

If $1 \le g < 4bB^W$, integrality gives $1 + g \le 4bB^W$ and $g < N_0$. The injective coefficients imply

$$|D_c(B)| < \frac{A_* B^{Q_*+1}}{B-1}, \qquad 2(1+g)^2 D_c(B) < \frac{32A_* b^2}{B-1} B^{K_*+1}.$$

Because $B = 8192A_*b^2$, the last prefactor is less than $1/128$. Together with $T_* < B^{Q_*+1}/(B-1)$ this gives $0 < J_c < B^{K_*+1}$. Likewise $(B-2)T_* < B^{Q_*+1} \le B^{K_*+1}$. Consequently

$$0 < S_c, T_c < B^{E_*}, \qquad S_c + T_c < 2B^{E_*} < L.$$

These estimates hold for arbitrary integer $b \ge 2$, not just powers of two.

Now suppose $b = 2^k$. Then $B = 2^\nu$, where $\nu = \kappa + 2k$. The binary mask M_c forces exactly

$$g \,\&\, M_c = 0 \quad \iff \quad g = \sum_i n_i B^{w_i}, \qquad 0 \le n_i < b.$$

Here & is used only in the proof. For such a code, write $h_e = [t^e](1 + \sum n_i t^{w_i})^2 D_c(t)$. Then

$$|h_e| \le A_*(1 + \textstyle\sum_i n_i)^2 < 1024A_*b^2 = B/8 < B/4.$$

The unnormalized digits of J_c are $2h_e + \varepsilon_e$, where $\varepsilon_e = 1$ at a target and 0 elsewhere. Starting with incoming carry $c_0 = 0$, normalized carries obey

$$c_{e+1} = \left\lfloor \frac{2h_e + \varepsilon_e + c_e}{B} \right\rfloor, \qquad c_e \in \{-1, 0\}.$$

At a target, its normalized digit avoids $B-2$ exactly when it is 0 or 1. If $h_e = 0$, the digit is $1+c_e \in \{0,1\}$. If $h_e > 0$, it is at least 2; if $h_e < 0$, a borrow normalizes it to a digit at least 2. Thus the target test is equivalent to $h_e = 0$, independently of incoming borrows. There is no need to fill every position with a dense $B/2$ offset. It follows, by splitting at N_0, that

$$S_c \,\&\, T_c = 0 \quad \iff \quad g \text{ codes a common zero of all } p_j.$$

Conversely every common zero has, for arbitrarily large $b = 2^{1+2r}$, such a code with $1 \le g < bB^W = c$.

7.3 The parity-tuned two-block threshold

Write $L = 2^\lambda$ after $b = 2^k$ has been established. The three disjoint blocks in R give

$$s_2(R) = 1 + s_2(S_c + T_c) + \lambda - s_2(S_c) = \lambda + s_2(T_c) + 1 - \tau_2(S_c, T_c),$$

where s_2 is binary digit sum and $\tau_2(S,T) = s_2(S) + s_2(T) - s_2(S+T) \ge 0$ counts binary carries. The strict inequality $S_c + T_c < L$ is essential. The 31 allowed low positions and the 22 target positions give

$$s_2(T_c) = (W+23)\nu - 31k - 22, \qquad \lambda = 1 + k + E_*\nu.$$

Therefore

$$\begin{aligned} \lambda + s_2(T_c) + 1 &= 5947\kappa + 11864k - 20 \\ &= 2\big(15\kappa/2 - 10 + 2966(\kappa + 2k)\big) = \log_2 N^2. \end{aligned}$$

Legendre's formula yields $v_2\binom{2R}{R} = s_2(R)$, hence

$$N^2 \mid \binom{2R}{R} \quad \iff \quad S_c \,\&\, T_c = 0.$$

The small factor $2b$ in L calibrates the threshold to a polynomial square without an extra variable, an extra target mask, or a cubic radix.

7.4 The interval guard and the Jones interface

If $\Delta > 0$ and $b \geq 2$, then $K \neq 0$ and $0 < g < c$. The identity $4g(c-g) = c^2 - (c-2g)^2 \leq c^2$ implies $4\delta^2 < K^2$. Along with $s \geq 1$, $\phi \geq 0$, and $w \neq 0$, this gives $C, Y > 0$ and $K > 0$. The usual square-gap argument forces $h \geq 1$: if $h \leq 0$, then $1 \leq K \leq R + 1 < P$; $K = 1$ would force $P - 1 \mid R$, which is impossible, while for $2 \leq K < P$,

$$(PK-1)^2 < (P^2-1)K^2 + 1 < (PK)^2.$$

Thus a_0 cannot be a square in that branch.

For precision, instantiate the free ceiling in `Jones1982.Nine.ratio_sound` by $G_{\mathrm{aux}} = 8gK^2 + 1$, rather than assuming its old margin condition. The positive integer $K^2 - 4\delta^2$ then gives

$$(K^2 - 4\delta^2)G_{\mathrm{aux}} - 8gK^2 > 0.$$

This ceiling is a proof-side abbreviation; the ratio registers do not otherwise depend on it.

Replace i by $|i|$, which changes no condition. To handle signed j, $u \mid t$ and $F \equiv 1 \pmod{J}$ imply $F \mid H - C$. Also $F \equiv 1 \pmod{D}$, $G \equiv 1 \pmod{D}$, and $G \equiv A \pmod{F}$ show that the positive integers D, F, I are pairwise coprime. Hence $a_1 = DFI$ square makes each a square. Put $d = G^2 - 1 > 0$ and $x = \sqrt{I}$. The norm-one unit $x + H\sqrt{d}$ is positive even if $H < 0$. A positive power of $G + \sqrt{d}$ is the identity modulo CF; multiplying by sufficiently large such powers gives a Pell solution $H' > B_1$ with $H' \equiv H \pmod{CF}$. Then $j' = (H' - B_1)/C > 0$, all square conditions remain true, and $u \mid t$ remains true. The literal positive Jones interface therefore applies and gives the power-of-two base and central-binomial condition required above.

For completeness, take any common source zero, choose $b = 2^{1+2r}$ large enough, and set $\epsilon = 2^r$. The direct source code has $32g < G_c$, so the literal ratio completeness theorem supplies its canonical witnesses. In `Ratio.lean`, `exists_B25core` sets

$$\omega = 2^{2R+1-k}, \qquad U = (N^2/b)2^{2R+1}.$$

Since $k = 1 + 2r$ is odd and $k < 2R + 1$, this ω is a positive square. No claim that the canonical s is a square is needed.

The same canonical construction has $\rho_0 = (U+1)^{2R}/U^R = Y + f$ with $0 \leq f < 2^{2R}/U = b/(2N^2)$ and $Y \geq 1$. The source's `ratio_core` gives

$$|C/K - \rho_0| \leq \frac{2(Y+f)}{Y(U+1)} < \frac{4}{U}, \qquad |C/K - Y| < \frac{b}{N^2}.$$

The last inequality uses $R \geq 8$. Since $c, b \leq N$ and $g(c-g) \geq 1$, it follows that

$$\frac{c^2(C - KY)^2}{K^2} < \frac{c^2 b^2}{N^4} \leq 1 \leq g(c-g),$$

so the very same canonical witnesses satisfy $\Delta > 0$. This is a cofinal square-base proof, not an unsupported restriction of an existing slack variable to squares.

7.5 Seeded square recovery, with all zero branches excluded

For every integer assignment, $J = 4A - 5 \neq 0$ and $F \neq 0$. Indeed $F = 0$ forces $A = 0$ and $E = \pm 1$, whereas E is then divisible by $J = -5$. Thus $u = JF \neq 0$ unconditionally. The polynomial

$$\Pi(x) = (x^3 - x - 1)(x^2 + 23)$$

has no rational root and has a root modulo every positive integer. For an odd prime other than 23, either its cubic has a simple root or that cubic is irreducible, whose Frobenius three-cycle makes its nonzero discriminant -23 a square; the quadratic then has a simple root. At 23 the cubic has the simple root 3. At 2, $-23 \equiv 1 \pmod 8$ has compatible square roots. Hensel lifting and CRT finish the assertion. Consequently $\Theta = u^5\Pi(t/u + r) \neq 0$ for every integer assignment. Moreover

$$\exists r \quad Zu^5 \mid \Theta \quad \Longleftrightarrow \quad Z \neq 0 \text{ and } u \mid t.$$

For the reverse implication, choose a root of Π modulo $|Z|$ after $u \mid t$. For the forward implication, if a prime has $v_p(u) = \alpha > \beta = v_p(t)$, the two factors of Θ have valuations 3β and 2β, contradicting $u^5 \mid \Theta$. If $Z = 0$, divisibility would demand $\Theta = 0$, already excluded.

At a zero of (14), $\Sigma = 0$ is impossible because then $\mathcal{F}_{13} = \Theta^4 \neq 0$. Therefore $\epsilon, w, i, z_1, z_2, z_3, a_0$ are all nonzero. Every $d_j \equiv 2 \pmod 3$ is nonzero, and every $B_j^* \equiv 1 \pmod 4$ is nonzero. Furthermore $B_j^* \neq 1, -1$, so $|B_j^*| \geq 3$ and $|C_*| > 1$ without any positivity hypothesis.

Let $q = X/\Sigma$. Equation (14) gives the exact identity

$$(q^2 + C_* - C_*^2 a_1)^2 = 4C_* q^2.$$

If $q = 0$, it would force $C_* a_1 = 1$, impossible since $|C_*| > 1$. Thus

$$r_0 = \frac{q^2 + C_* - C_*^2 a_1}{2q}, \qquad r_1 = \frac{q - r_0}{C_*}$$

are rational and satisfy $r_0^2 = C_*$, $r_1^2 = a_1$. Rational square roots of integers are integers. Hence C_*, a_1 are integer squares, $q \in \mathbb{Z}$, and $\Sigma \mid \Theta$. This proof covers negative radicands and all possible cancellations without an implicit positivity or distinct-square-class assumption.

Now $\gcd(B_j^*, C_{j-1}) = 1$, so the nonzero factors a_0, B_1^*, B_2^*, B_3^* are pairwise coprime. Since their product C_* is a square, every absolute factor is a square. Each $B_j^* \equiv 1 \pmod 4$ is therefore positive, because a negative square cannot be 1 modulo 4. The sign of the product forces $a_0 > 0$, and a_0 is a square too. All prefixes are now positive. A negative d_j would make $B_j^* \leq -3$, so all $d_j > 0$, yielding $s \geq 1$, $\phi \geq 0$, and $\Delta > 0$. This argument does not assert that $a_0 \equiv 1 \pmod 4$; that false shortcut would invalidate the seeded construction.

Conversely, suppose a_0 is a positive square and these three sign conditions hold. Each d_j is positive and nonsquare since it is 2 modulo 3. Inductively C_{j-1} is a positive square, so $d_j C_{j-1}$ is positive nonsquare. A nontrivial Pell solution $x^2 - d_j C_{j-1} y^2 = 1$ can be chosen with y even; set $z_j = y/2 \neq 0$. Then $B_j^* = x^2$ and C_j is a positive square. Given also a_1 square and $u \mid t$, the intersective lemma chooses r with $\Sigma \mid \Theta$, and an integer choice of v makes $X/\Sigma = \sqrt{C_*} + C_*\sqrt{a_1}$. Thus (14) vanishes. Combining both directions proves the asserted equivalence to the original source and hence to a finite ZFC refutation.

7.6 Exact degree, circuit accounting, and provenance

For the seeded equation the relevant exact register degrees are

	R	N	a_0	a_1	C_*	Σ
deg	45932	11864	652210	4363426	5592732	6692101

and $\deg X = 6692102$, $\deg A_0 = 18976934$, $\deg A_1 = 28933092$. The last dominates X^2 and A_0, so the output's leading term is A_1^2. The symbolic diagonal certificate has leading coefficient

$$356960\, a^{14465376}\eta^{1888} \pmod{1000003},$$

which is nonzero because a and η are powers of two. The circuit upper bound and this surviving diagonal coefficient prove $\deg \mathcal{F}_{13} = 2(28933092) = 57866184$ exactly, not merely at most.

The metric counts every live binary addition, subtraction and multiplication, including scalar multiplications and every gate in fixed exponentiation. Fixed coefficient leaves and reuse are free; coefficient bit size is a separate measure.

The unseeded three-root baseline, also included, uses the same source, starts $C_0 = 1$, omits a_0 from Z, and combines the square conditions C_*, a_0, a_1 with the established separating weights $1, C_*, C_*(P^2 + K^2 + 2)$. It has 770 operations and degree 71382096. The seeded improvement saves eleven operations and reduces this degree by 18.93% without changing the thirteen unknowns. For completeness, if $W_2 = C_*(P^2 + K^2 + 2)$, its radicands are $A_0 = \Sigma^2 C_*$, $A_1 = \Sigma^2 C_*^2 a_0$ and $A_2 = \Sigma^2 W_2^2 a_1$. It uses

$$V = X^2 + A_2 - A_0 - A_1, \quad W' = V^2 + 4X^2 A_2 - 4A_0 A_1, \quad (W')^2 - 16V^2 X^2 A_2.$$

This is the product over all eight signs of $X - \Sigma(\pm\sqrt{C_*} \pm C_*\sqrt{a_0} \pm W_2\sqrt{a_1})$. The inequalities $|C_*| > \sqrt{|C_*|}$ and $\sqrt{|a_0|} \le P^2 + K^2 + 1$ make each weighted nonzero radical larger in complex absolute value than the sum of all preceding radicals. Thus effective sign patterns are distinct, also when a radicand is zero. A rational signed sum is fixed by every multiquadratic automorphism; uniqueness forces each nonzero root rational and hence integral. This recovers all three square conditions and the merger divisibility. At $\Sigma = 0$ its output is $\Theta^8 \neq 0$. The unseeded Pell stack and all remaining source and ratio steps are the same as proved above.

Run `python build_jones.py` and `python check_jones.py`. The checker authenticates the fixed payloads and pinned sources, independently replays the literal registers over three prime fields (including signed and zero inputs), verifies the formal norm identities, checks 136000 signed seeded stacks, and exercises nonzero merger and sparse-borrow branches. These finite tests corroborate the universal proofs, not replace them. All imported source paths, Git blob IDs, and SHA256 values are in `SOURCE_PINS.json`, at repository commit `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. The inherited compiler and Jones ratio theorems are explicitly identified; no newly compiled kernel certificate for this concrete equation is claimed.

8 Eleven positive unknowns with denominator reduced absorption

The new equation has eleven positive unknowns, exact degree 958, and 117 charged binary arithmetic operations. A second equation trades four degrees for two operations: degree 962 and 115 operations. The full carrier's two-equation system has degrees $(2, 956)$ and uses 112 operations after the standard positive transport shift. These improve both pinned eleven-positive frontiers, 1154/123 and 1140/129. They do not dominate the twelve-positive degree-901 construction. The argument uses no Sun representation theorem.

Two changes make this possible. First, the multiplier-one norm from source44 remains sound without the deleted index equation: its main-root bound needs $d(I\sqrt{G} - c\sqrt{D}) > b + c\sqrt{D}$, not the stronger $I\sqrt{G} - c\sqrt{D} > 1$. Second, quotient absorption needs denominator Y, not $E = XY$. The short Pell-index argument recovers the full E-divisibility after the weaker Y-divisibility has done its job.

8.1 Explicit equation

Take precisely the positive integer unknowns

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, t.$$

Fix the complete authentic normalized blank-K215 coefficient tuple (β, K_0, M_C, M_F), with M_F already the shifted high mask. The delivered fixed instance is the materialized Wang943 121×8801 tuple with

$\beta = 2^{1064921} - 1$; its four payloads and hashes are included. The proof is uniform in any whole authentic tuple. Set

$$\begin{gathered}
q = \beta J + 1, \quad s_0 = s + 3q, \quad X = wq, \quad Y = s_0 q^3, \quad E = XY, \\
k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \\
\Delta = a^2 + H = (a+2)^2 - 1, \quad \ell = EkY, \quad A = 1 + \ell(\ell + k), \\
G = 1 + \Delta c^2, \quad b_0 = X + ac, \quad C_w = q - F - \alpha, \\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
N_t = (K_0 + w)C_w + q - F - z(q-1), \\
I = ic^2, \quad D = 1 + \Delta I^2, \quad Q = \Delta(D-1) = (\Delta I)^2, \\
d = Q - 1, \quad v = cT, \quad b = c + RD, \\
e = Q(Dv^2 + b^2) - 1, \quad B_* = 2Qvb, \quad K = IT + 1, \\
L_0 = 1 + HK, \quad S = HY, \quad L = YL_0.
\end{gathered}$$

The computed register K is not the fixed coefficient K_0. For the full carrier put $\delta = (a+2)c - \ell$; for the shorter one put $\delta = ac - \ell$. In either case define

$$W = Y\big[b_0 + H(t + b + \delta)\big] + H(k - R - 1)$$

and the integer polynomial

$$N = \prod_{\substack{g=\pm\sqrt{G} \\ \tau=\pm\sqrt{A} \\ f=\pm\sqrt{D}}} \left[d(W - Lg - S\tau)^2 - S^2 e + S^2 B_* f\right]. \tag{15}$$

The formal signs define a conjugate product, not independently assumed Galois automorphisms. Every other symbol is a polynomial abbreviation, not another unknown. The proposed single equation is

$$\boxed{N_t(1+N) - 1 = 0}. \tag{16}$$

The two-equation presentation is $N_t = 1$, $N = 0$.

For an explicit division-free scalar evaluator, set

$$\begin{aligned}
d_n &= dS^2 A, & H_Q &= S^2 Q, & H_P &= H_Q D v^2, & H_U &= H_Q b^2, \\
H_e &= H_P + H_U - S^2, & j &= 4H_P H_U, & & & & \\
\mathcal{T} &= 2d(W^2 + L^2 G), & \mathcal{K} &= [d(W^2 - L^2 G)]^2, & & & & \\
A_* &= 2(d_n + H_e), & C_* &= (d_n - H_e)^2 - j, & & & & \\
u &= \mathcal{T} - A_*, & r &= C_* - \mathcal{K}, & c_* &= 16 d_n j, & & \\
V_* &= c_* - u(u\mathcal{T} + 2r), & U_* &= r^2 - u^2 \mathcal{K}, & & & & \\
N &= \mathcal{K} V_*^2 - U_*(\mathcal{T} V_* - U_*). & & & & & &
\end{aligned}$$

This is a universal polynomial identity, including degenerate radicands. The emitted JSON circuits expand every square and count every binary gate.

8.2 Weighted main-root separation

Let $x_0 = d(W^2 + L^2 G)$. Since $\mathcal{T} = 2x_0$ and $\mathcal{K} \equiv x_0^2 \pmod 4$, the norm obeys $N \equiv (U_* - x_0 V_*)^2 \pmod 4$. Thus a root of (16) forces $N = 0$, $N_t = 1$: the alternative integer unit $1 + N = -1$ would make $N = -2$. The inherited transport interface then gives

$$q \geq 16, \quad C_w > 0, \quad 3q < R < q^4 - 2.$$

Our positive shift implies

$$X \geq 16, \quad Y > 3q^4 > 3R, \quad E > R+1, \quad k \geq 2, \quad kY < c < 2kY, \quad c > 96, \quad a+2 < 2E.$$

No assertion $k > E$ has been used.

For positive real $g = \sqrt{G}$, $f = \sqrt{D}$, $\tau = \sqrt{A}$, direct comparison yields

$$1 < A < \Delta c^2 \leq \Delta I = \sqrt{Q}, \qquad g > \tau, \quad g < 2Ec, \quad f < 2EI.$$

The exact identity $(Ig)^2 - (cf)^2 = I^2 - c^2$ and $I \geq c^2$ give

$$Ig - cf > \frac{I}{8Ec}.$$

Moreover $b + cf < 6E^3I^2$ and $d > \frac{1}{2}E^4I^2$, so

$$d(Ig - cf) > \frac{E^3I^3}{16c} \geq \frac{E^3I^2c}{16} > 6E^3I^2 > b + cf. \tag{17}$$

This is the new estimate that removes dependence on the missing index quotient.

Every root of the inner norm has a coherent representation

$$x = \tau' + y, \quad (\tau')^2 = A, \quad (f')^2 = D, \quad dy^2 = Q(vf' - b)^2 - 1.$$

These roots may be complex. With $Z = b + vf > 1$,

$$|y| < Z + Z/d, \qquad Z/d \leq T(b+cf)/d < T(Ig - cf)$$

by (17). Hence $|x| < b + (IT+1)g = b + Kg$. But the negative main sign gives

$$\frac{W + Lg}{S} = b + t + \delta + \frac{k - R - 1}{Y} + \frac{b_0}{H} + \left(K + \frac{1}{H}\right) g > b + Kg,$$

since $t + (k - R - 1)/Y > 0$. This is impossible. Quadratic conjugation therefore forces the main root g to be a positive integer, and $x = (W - Lg)/S$ is rational.

8.3 Integral inner roots

Set $C(x) = d(x^2 + A) - e$. The inner norm is

$$F(x) = \left[C(x)^2 + DB_*^2 - 4d^2x^2A\right]^2 - 4C(x)^2DB_*^2.$$

If $C(x) = F(x) = 0$, substitution gives $d^2A^2 - deA + Q^2v^2b^2D = 0$, so $Q \mid A(A-1)$, impossible because $1 < A < \sqrt{Q}$. Also $F(0) = 0$ would imply $Q \mid (A-1)^2$, again impossible. The rational expressions

$$f' = -\frac{C(x)^2 + DB_*^2 - 4d^2x^2A}{2C(x)B_*}, \qquad \tau' = \frac{C(x) + B_*f'}{2dx}, \qquad y = x - \tau'$$

therefore have justified denominators and satisfy $(f')^2 = D$, $(\tau')^2 = A$ and the auxiliary equation. Thus f', τ' are integers. The inherited rational Pell saturation lemma,

$$m^2V^2 - (m^2-1)y^2 = 1, \quad m, V \in \mathbb{Z},\ m \geq 2, \quad y \in \mathbb{Q} \quad \Longrightarrow \quad y \in \mathbb{Z},$$

with $m = \Delta I$ and $V = vf' - b$ makes y, hence x, integral. A proof of the lemma is included in the companion proof file; it uses abscissa and ordinate divisibility in the integer-coordinate Pell group. Reducing $W = Lg + Sx$ modulo Y now yields

$$Y \mid H(k - R - 1). \tag{18}$$

8.4 Short indices restore full divisibility

Write $A_0 = a+2$ and let χ_{A_0}, ψ_{A_0} be its Pell sequences. The main and strong roots give positive indices p, m with

$$g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \quad |f'| = \chi_{A_0}(m), \quad I = \psi_{A_0}(m).$$

Strong divisibility and $I = ic^2$ give $p \mid m$; composition modulo c then gives $c \mid m/p$, so $m \geq pc > 2p$. Also $p < c/2$ and $R < c/2$. Source38's signed auxiliary-index argument, applied before (18), yields $R \equiv \pm p \pmod{c}$, hence $p = R$. For clarity, it classifies the auxiliary Pell solution at ΔI with odd index j_0. Reducing its odd-index polynomial modulo $|f'|$ gives $j_0 \equiv \pm p \pmod{m}$, while reduction modulo c gives $j_0 \equiv \pm R \pmod{c}$. The full recurrence argument is reproduced in the companion proof file, including the signed branches.

The square A has fundamental integer-coordinate unit $(2EY+1, 2)$, so for some $n \geq 1$,

$$|\tau'| = \chi_{2EY+1}(n), \qquad k = 2\psi_{2EY+1}(n).$$

Monotonicity and $2EY + 1 > A_0$, $c > k$ imply $n < p = R$. Since $2EY + 1 \equiv 1 \pmod{E}$,

$$k \equiv 2n \pmod{E}.$$

Now $\gcd(Y, H) = \gcd(Y, 3) \leq 3$. By (18),

$$2n \equiv R + 1 \pmod{\mu}, \qquad \mu = Y/\gcd(Y, H) > R.$$

But $|2n - R - 1| < R$, so $2n = R + 1$. It follows that $E \mid k - R - 1$ after all, and

$$h = (k - R - 1)/E > 0.$$

Dividing the carrier by Y gives

$$b_0 + H(t + b + \delta + hX) = L_0 g + H(\tau' + y).$$

Thus $g \equiv b_0 \pmod{H}$. The recovered $\gamma = (g - b_0)/H$ is positive: $n \geq 2$ gives $k > X$, hence $c > X$, and $g > (a+1)c$. All normalized compiler equations are restored. Absolute values of τ', y are harmless; simultaneous sign reversal of f', T is covered by the inherited signed-quotient interface. This proves soundness through the calibrated blank-K215 halt equivalence.

8.5 Completeness and exact degree

For a canonical compiler witness, $X = 2^R$, $Y > 2^R$, and $R > 3q$ ensure $s_{\text{old}} = Y/q^3 > 3q$. Thus $s = s_{\text{old}} - 3q > 0$. Choose positive canonical roots and define

$$t = \gamma + Kg + \tau + y - b - \delta - hX.$$

Since $0 < \delta < g$, $y \geq b - vf$, and $\tau > \ell = EkY > k/Y > hX$, one has

$$t > \gamma + \tau + T(Ig - cf) - hX > 0.$$

This produces the coherent zero factor and proves completeness. The transport substitution $z_{\text{old}} = z+1$ replaces $N_t = 1$ by $(K_0+w)C_w = F+z(q-1)$ and saves one operation; canonical witnesses have $z_{\text{old}} \geq 2$.

Let N_{12} be the earlier multiplier-one norm. The exact localized identity is

$$N = Y^{16} N_{12}\left(s_{\text{old}} = s + 3q, t_{\text{old}} = t + \frac{k - R - 1}{Y}\right).$$

Each of the eight factors contributes Y^2, and the t_{old} degree is at most 16, so denominators clear. The rational shift has degree 0, below the degree-1 supplied carrier. It cannot change the highest form. The invertible highest-linear substitution $s_{\mathrm{old}} = s + 3\beta J$ preserves its nonvanishing. More explicitly, the rational carrier perturbation changes an unscaled factor only in degrees at most 96 and 58, below its smallest established factor degree 97. The earlier exact norm degrees 892 and 896 consequently become 956 and 960. The transport factor has exact degree 2, giving 958 and 962.

The full highest form is Y_0^{16} times

$$-2^{14}H_0^{16}d_0^8B^8V^4\ell_0^2k_0Y_0^2\left(2t+k_0-\frac{iTc_0}{E_0}\right),$$

where

$$q_0=\beta J,\quad Y_0=(s+3q_0)q_0^3,\quad X_0=wq_0,\quad E_0=X_0Y_0,\quad c_0=k_0Y_0,\quad k_0=\eta+\zeta,$$
$$I_0=ic_0^2,\quad H_0=4E_0,\quad \ell_0=E_0c_0,\quad d_0=E_0^4I_0^2,$$
$$B=q_0^3(q_0-F)E_0^2I_0^2,\qquad V=I_0T\ell_0.$$

Its t coefficient is nonzero for every authentic $\beta > 0$; the displayed denominator cancels. Multiplying by $w(q_0 - F - \alpha) - zq_0$ gives the single equation's highest form. The companion proof gives the short carrier's form as well.

8.6 Verification and logical boundary

The included builder audits all gates and coefficient hashes. The independent checker passes 24 exact and 60 modular formal conjugate products per carrier, three complete finite-field polynomial rays with both illustrative and actual fixed coefficients, all stated leading forms, the transport shift, and 288 pre-index bound tests. The tests include $k = 2$ and arbitrarily large X, where the old unweighted gap shortcut is unavailable. Both normal Python and optimized Python pass. Finite checks validate the implementation, not the unbounded compiler.

The baseline is ProveIt commit `124c07c0b77a`, specifically source38's index reconstruction and source44's multiplier-one norm. Complete source pins, copied proof dependencies, payloads, circuits, and the detailed mathematical proof accompany this section. The resulting theorem is that ZFC proves the absence of positive solutions equivalent to Con(ZFC). No consistency decision, global minimality theorem, independent human review, or new kernel-checked end-to-end certificate is asserted.

9 A 72-operation, degree-80 relaxed-kernel equation

The construction here has fifteen strictly positive integer unknowns, exact total degree 80, and 72 charged binary arithmetic operations. It saves one multiplication against the pinned degree-80 normalized P_{73} evaluator. This is a local tradeoff, not a global minimality claim: the inherited degree-155 equation still uses only 67 operations. The relaxed norm itself is prior art. Historical package 10 already has a degree-56, 74-operation equation using it; our companion P_{74} merely recovers that pair by a different grouping.

9.1 The explicit equation and its fixed coefficients

The witnesses, all in $\mathbb{Z}_{>0}$, are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma.$$

The four fixed ports are the *complete* materialized Wang943 121-by-8801 tuple, with

$$B=2^{1064921},\quad \beta=B-1,\quad K_0=D_C+BD_R,\quad M_C,\quad M_F=M_{F,\mathrm{native}}+B-1.$$

The files `COEFFICIENT_BINDING.json` and `coefficients/` specify their exact bytes and hashes. The complete native adapter proof is retained as `upstream_verified/COEFFICIENT_PROOF.md`. No component is taken from the different complement-packed coefficient tuple.

Define polynomial abbreviations, not additional unknowns:

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, \\ Y &= sq^3, & E &= XY, & k &= \eta + \zeta, \\ c &= kY + \eta, & a &= E + Y, & A &= a + 2, \\ H &= 4a + 3, & \Delta &= a^2 + H = A^2 - 1, & D &= X + ac + \gamma H. \end{aligned}$$

The remaining abbreviations are

$$\begin{aligned} G &= (q-1)(q-F) + \alpha, & R &= G(q^2-1) + (M_C + qM_F)J, \\ V &= c(Tf - 1) - Rf^2, & I &= ic^2, \qquad K_{\mathrm{aux}} = \Delta(f^2 - 1). \end{aligned}$$

Set

$$\begin{aligned} N_0 &= \tau^2 - (XY^2)(XY^2+1)k^2, & N_1 &= D^2 - \Delta c^2, \\ N_a &= K_{\mathrm{aux}}(V^2 - y^2) + y^2, & N_k &= k - hE - R, \\ N_t &= (K_0 + w)C + (q - F) - z(q-1), & S &= I^2 - K_{\mathrm{aux}}. \end{aligned}$$

Our equation is

$$\boxed{(N_0N_1 - 1)^2 + (N_aN_kN_t - 1)^2 + S^2 = 0.} \tag{19}$$

The companion is

$$(N_0 - 1)^2 + (N_a - 1)^2 + S^2 + (N_1N_kN_t - 1)^2 = 0. \tag{20}$$

The literal files `p72_relaxed.json` and `p74_relaxed.json` contain the full division-free evaluators. Only fixed nonnegative exponents occur.

9.2 What must be proved after relaxing the norm

The new condition is

$$(ic^2)^2 = \Delta(f^2 - 1),$$

not the normalized condition $f^2 - \Delta(ic^2)^2 = 1$. These individual equations are not equivalent at the same coordinates. In completeness, the old multiplier is scaled by Δ; in soundness, a rank argument must first recover integral Pell coordinates. Consequently the existing Lean same-coordinate theorem for normalized P_{75} cannot be cited as a certificate for this polynomial.

At a zero of (19), $K_{\mathrm{aux}} = I^2$ and each of N_0, N_1, N_a, N_k, N_t is an integer unit. The negative-Pell descent for $v(v+1)$, with $v = XY^2 > 1$, excludes $N_0 = -1$. Since $\Delta = (a+2)^2 - 1 \equiv 0$ or $3 \pmod 4$, $N_1 \neq -1$. Since K_{aux} is a square, $N_a \equiv y^2$ or $V^2 \pmod 4$, so $N_a \neq -1$. Thus

$$N_0 = N_1 = N_a = 1, \qquad N_k = \epsilon, \quad N_t = \nu, \qquad \epsilon\nu = 1, \quad \epsilon, \nu \in \{-1, 1\}. \tag{21}$$

The same conclusion follows from (20). Neither remaining sign is assumed positive at this point.

Weak transport implies $C \geq 0$, not yet $C > 0$. The authentic mask ranges then give

$$q(q^2-1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \qquad R - 2 > 3q + 1, \qquad E > R + 2, \qquad a > R + 2.$$

Let $P_0 = 2XY^2 + 1$. Ordinary Pell classification yields

$$\tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \qquad D = \chi_A(p), \quad c = \psi_A(p).$$

Since $P_0 \equiv 1 \pmod{E}$, $2n = R + \epsilon + bE$ for some $b \geq 0$. Thus $n \geq 24$. Moreover $P_0 > A$ and $c > k$ give $p > n$; Pell growth gives

$$c > A\Delta^2, \qquad c > 2p, \qquad c > 2R. \tag{22}$$

No auxiliary index or decoded tableau is used for these bounds.

9.3 The inherited relaxed-rank lemma, proved explicitly

Suppose $c = \psi_A(p) > A\Delta^2$ and $(ic^2)^2 = \Delta(f^2 - 1)$. Then there is $m > 0$ with

$$f = \chi_A(m), \qquad p \mid m, \qquad c \mid m, \qquad ic^2 = \Delta\psi_A(m). \tag{23}$$

Here is the full rank argument. Write $\Delta = d_0 s_0^2$ with $d_0 > 1$ squarefree and choose the least positive integer-coefficient unit $F + y_0\sqrt{d_0}$. By division by powers of this least unit, every positive integer-coefficient norm-one unit is its power. For some $e \geq 1$,

$$A = \chi_F(e), \quad S_0 = \psi_F(e), \quad s_0 = y_0 S_0, \quad \Delta = (F^2 - 1)S_0^2, \quad \psi_F(ep) = S_0 c.$$

The relaxed equation implies $s_0 \mid ic^2$ and $d_0 \mid ic^2/s_0$, so it too determines an integral norm-one unit. For some $t > 0$,

$$f = \chi_F(t), \qquad ic^2 = (F^2 - 1)S_0\psi_F(t).$$

Put $L = (F^2 - 1)S_0 \leq \Delta$ and $g = \gcd(t, ep)$. Strong divisibility of the Pell sequence gives

$$c\gcd(c, \Delta) = \gcd(c^2, L\psi_F(ep)) \mid L\psi_F(g), \qquad \psi_F(g) \geq c/\Delta.$$

If $g < ep$, then $2g \leq ep$ and doubling gives

$$2\psi_F(g)^2 < \psi_F(2g) \leq \psi_F(ep) = S_0 c < Ac,$$

contrary to $c^2/\Delta^2 > Ac$. Hence $ep \mid t$. Write $t = em$; composition proves $p \mid m$ and the first and last claims in (23).

For the remaining divisibility, write $m = pb$. The binomial expansion gives

$$\psi_A(pb)/c \equiv bD^{b-1} \pmod{c}.$$

Since $c^2 \mid \Delta\psi_A(m)$ and $\gcd(D, c) = 1$, we have $c \mid \Delta b$. Also $c \equiv pA^{p-1} \pmod{\Delta}$, so $g_0 = \gcd(c, \Delta) = \gcd(p, \Delta)$ divides p. Thus $c/g_0 \mid b$ and $c \mid pb = m$, as required. This proof does not assert the stronger, generally unavailable condition $c^2 \mid \psi_A(m)$.

9.4 Quotient elimination is still sound

The rank lemma must be used *before* restoring the two deleted quotients. Set $\rho = ic^2$. Since $p \mid m$, we have

$$c^2 \mid f^2 - 1, \quad f^2 \geq 1 + \Delta c^2, \quad f > 2c, \quad \rho^2 = \Delta(f^2 - 1), \quad c \mid \rho, \quad m \geq c > 2p.$$

As $R < a < \Delta$,

$$\rho^2 - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c \geq f^2 - \Delta - c > 0.$$

The auxiliary equation is $\rho^2 V^2 - (\rho^2 - 1)y^2 = 1$. If $|V| > 1$, it implies $|V| \geq 2\rho^2 - 1$. But $T > 0$ gives $V > -(Rf^2 + c) > -\rho^2$, excluding negative values with $|V| > 1$. The cases $V = \pm 1$ contradict $V \equiv -c \pmod{f}$ and $f > 2c > 4$. Thus $V > 0$.

Now the proof-only expressions

$$o = cT - Rf, \qquad j = Tf - R(f^2 - 1)/c - 1$$

are integers, and $of = V + c > 0$, $jc = V + R > 0$. Consequently $o, j > 0$ and

$$V = of - c = jc - R.$$

There is no division in the polynomial evaluator. In particular, the old normalized formula for j in terms of $\Delta i^2 c^3$ must not be used with the new multiplier.

The retained signed auxiliary step-down applies with exactly these hypotheses and (22). To expose its interface, write $\rho V = \chi_\rho(\ell)$ with $\ell = 2u + 1$ odd and $\chi_Z(2u + 1) = ZQ_u(Z^2)$. The identities

$$Q_u(1 - A^2) = (-1)^u \psi_A(\ell), \qquad Q_u(0) = (-1)^u \ell$$

give $(-1)^u \psi_A(\ell) \equiv -c \pmod f$ and $(-1)^u \ell \equiv -R \pmod c$. Squaring the first and applying the retained plus-congruence step-down to $\chi_A(2\ell) \equiv \chi_A(2p) \pmod{\chi_A(m)}$ gives $\ell \equiv \pm p \pmod m$, hence $R \equiv \pm p \pmod c$. The half-interval bounds force $p = R$.

Returning to $2n = R + \epsilon + bE$, the bound $E > R + 2$ rules out $b > 0$. If $\epsilon = -1$, then $p = 2n + 1$, and duplication with $\chi_A(2) > P_0$ gives $c > Ak > k(Y + 1)$, contradicting $kY < c < k(Y + 1)$. Therefore $N_k = N_t = 1$ and only now $C > 0$. The retained fixed-minus parity theorem yields $R \equiv 3 \pmod 4$. All remaining lower-ratio, half-binomial, mask and native-compiler steps then have their original hypotheses. Exact sources and full proof-order details are in `PROOF.md` and the pinned `INHERITED_PELL_AUDIT.md`.

9.5 Completeness and the logical endpoint

From a finite ZFC refutation, the inherited normalized compiler supplies a positive tuple for this entire selected coefficient tuple. Keep all its witnesses except replace i_{old} by $i_{\text{new}} = \Delta i_{\text{old}}$. Then

$$(i_{\text{new}} c^2)^2 = \Delta(f^2 - 1),$$

while K_{aux}, V and all outer factors are unchanged. Every required factor is $+1$, so both displayed equations vanish. Conversely the preceding soundness proof recovers a calibrated finite blank-K215 computation. Hence, in ordinary mathematics formalizable in ZFC,

$$\text{ZFC} \vdash \left(\neg \exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{72}(\mathbf{x}) = 0\right) \longleftrightarrow \text{Con(ZFC)}.$$

This is not a proof of consistency and not a new end-to-end proof-assistant certificate. It uses the inherited compiler and literal K215 calibration.

9.6 Exact degree, operation counts, and historical comparison

The kernel has 61 live producers: 34 multiplications and 27 additions/subtractions. Equation (19) adds three products, three residual subtractions, three squares and two additions: $72 = 40M + 32A$. Its residual degrees are $40, 37, 22$. The companion has $74 = 40M + 34A$, with degrees $22, 28, 22, 27$.

The main norm's cancellation is an ordinary polynomial identity:

$$(ac + Z)^2 - (a^2 + H)c^2 = 2acZ + Z^2 - Hc^2, \qquad Z = X + \gamma H.$$

The independent auditor expands that cone exactly before propagating leading homogeneous forms. The two polynomial leaders are respectively

$$64\gamma^2 w^8 s^{14} (\eta + \zeta)^6 (\beta J)^{50}, \qquad w^4 s^8 T^4 f^8 (\eta + \zeta)^4 (\beta J)^{28}.$$

Their nonzero coefficients $64\beta^{50}$ and β^{28} prove exact degrees 80 and 56 for the actual positive fixed β.

The historical pairs (degree, operations) $(99, 69), (93, 70), (83, 71), (56, 74), (44, 78)$ remain relevant. The present $(80, 72)$ point does not dominate $(83, 71)$ or $(56, 74)$; it lies between them. In particular, $(56, 74)$ is *not* a new record here. The prior relaxed kernel and its historical Q_{56} are explicitly retained for comparison. The reproducible standard-library audit verifies counts, liveness, literal sources, complete payloads and exact degree, but does not itself certify the unbounded compiler semantics.

10 Proof status and newer repository developments

The repository has made important formal progress independent of these explicit reductions. At the post-baseline snapshot `59e9d4d22b34abdcc27f378796b9bfa28c6f1cc2`, its status document reports a closed executable nine-variable expression from the concrete refutation enumerator, with checked external Lean equivalences `ExplicitZFC.noSyntaxZero_iff_conZFC` and `noNaturalZero_iff_conZFC`. Its numerical coefficient payload is not materialized. The same document still marks the P67 compiler proof in Lean and an object-language ZFC derivation as open. We inspected these status statements, but did not rebuild that Lean development.

These are different constructions and proof layers. Our finite, materialized circuits do not inherit a kernel certificate merely because another executable expression has a Lean theorem. Conversely, it would now be incorrect to describe all repository coefficient recipes as noncomputable choices. The report preserves this distinction and includes the reviewed status document as provenance.

11 Validation and reproducible delivery

The root `verify.py` authenticates the release, copies it to an isolated temporary directory, and executes 21 finite replay stages. It never modifies the sealed originals. Python 3.10 or later, SymPy and NumPy are required; the versions used for the recorded run are Python 3.12.14, SymPy 1.14.0 and NumPy 2.3.5. The exact tested package versions are listed in `requirements.txt`.

The checks are deliberately separated by logical strength:

- Exact source reconstruction authenticates all four complete coefficient ports, positive-to-natural coordinate shifts, and each used quadratic coefficient. The original parent31 transport convention is selected explicitly; the related package45 row is not silently substituted.
- Exhaustive finite certificates verify all 71,808 monomial interactions and 22 target identities. Eleven deliberate mutations of the selective certificate or its pinned source data are rejected.
- Algebraic audits check polynomial identities in local register rings, independent literal evaluations, formal conjugate products and cancellation-aware degree certificates. A nonzero leading residue proves that a degree upper bound is attained; random evaluation alone would not.
- Independent finite controls cover binary overflows, signed borrows, parity-tuned thresholds, and seeded norms with negative as well as positive inputs. Their proofs are universal arguments in the text; the finite ranges are diagnostic corroboration, not unbounded proofs.
- Byte-for-byte rebuilt-file comparison detects changes to delivered circuits. Three separate release corruption controls test only integrity; they are not advertised as three independent mathematical counterexample searches.

The exact stage list, output list, source digests and replay receipt are machine-readable. The root checker is run without `-O`; individual checker receipts also record optimized-Python replays. No received upstream proof script is used as a substitute for a mathematical argument, and no Lean or Rocq endpoint build is claimed for these new equations.

The root `build_report.sh` compiles this editable TeX source twice on a standard TeX Live installation. The release also includes component proofs, their exact polynomial DAGs, coefficient recipes and inherited notices. No network request is needed for finite reproduction. The unbounded imported results have immutable source references; reproducing the entire upstream theorem-prover development is outside this finite replay contract.

12 Limits and next mathematical targets

The smaller selective code is a concrete certificate, not a minimum among all labelings. The native coefficient tuple is kept coherent; replacing one port in isolation is forbidden. A smaller published tuple could be substituted only with its complete adapter and a newly verified bound on every derived coefficient. We do not hide that cost by calling all large integers unit-size data.

The attempt to absorb the remaining transport quotient is not promoted to a ten-positive-variable theorem. Removing the transport equation loses the early positivity and size bounds; a proposed rational-index repair leaves a modular branch that does not force the required original index congruence. The archive records the obstruction separately from validated constructions. No search for an actual consistency refutation is performed.

The most promising next proof-assistant target is the general selective extraction and signed-borrow theorem: it has a small finite certificate interface and is independent of the set-theoretic calibration. The calibrated two-block carry lemma is similarly elementary. Formalizing these reusable interfaces would be more valuable than treating finite sample tests as if they certified the complete Diophantine reduction.

Attribution and references

The classical Pell, Jones, and Sun representation results are inherited, not claimed as new. Source38 supplies direct quadratic batching and index absorption; sources39,40,43 and45 develop direct or multi-output quadratic encoding; source44 supplies the multiplier-one norm and earlier signed Jones guards. The new finite labeling, overlapping-target certificate, calibrated sparse carry adapter, seeded guard and denominator reduction are described with their specific proof obligations above. Historical short-form comparisons are checked against earlier waves as well as the newer Lean-local frontier. Source URLs and exact file digests are in the component ledgers and root dependency map. AI-assisted internal reviews are not represented as independent human endorsement.